



Uncertainty-aware disentangled representation learning for multimodal fake news detection

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ABSTRACT

The proliferation of fake news on social media platforms poses serious societal risks, such as eroding public trust, inciting panic, and influencing policy decisions. While automated multimodal fake news detection has emerged as a promising approach, existing methods face three critical limitations: (1) they often fail to capture uncertainty within multimodal data, (2) they struggle with modality heterogeneity, and (3) they lack a balanced focus on both modality-private veracity and cross-modal inconsistencies. In this work, we propose a novel Uncertainty-aware Disentangled Representation Learning (UDRL) framework that addresses these limitations in three key ways. First, our probabilistic representation module models multimodal information as Gaussian distributions, effectively capturing uncertainty and ambiguity. Second, we introduce a disentangled representation learning framework that separates shared and private modality information, enhancing robustness and discrimination. Finally, our uncertainty-aware fusion module dynamically adjusts modality importance based on uncertainty, facilitating more accurate cross-modal interactions. Experimental results on three benchmark datasets demonstrate that UDRL achieves competitive and consistent performance across datasets, validating its effectiveness in multimodal fake news detection.

1. Introduction

In recent years, the rapid development of social media platforms such as Facebook, Twitter, and Instagram has significantly transformed the way information is disseminated. However, this evolution has also led to the proliferation of fake news. Fake news spreads quickly on social media, undermining public trust in the media, causing societal confusion, and misleading important decisions (Bazmi, Asadpour, Shakery, & Maazallahi, 2024). Manual detection of fake news, on the other hand, is time-consuming, labor-intensive, and often prone to mistakes due to human bias and limitations. Therefore, study on automated fake news detection has become an urgent and critical task.

With the rapid advancements in multimedia technology, fake news detection has evolved from early content-based methods that focused solely on textual modalities – such as linguistic features, sentiment analysis, and topic modeling – to approaches that leverage multiple modalities, including text, image, and video. Multimodal fake news detection has garnered significant attention in recent years due to its ability to incorporate richer information and enhance detection performance. As illustrated in Fig. 1, existing multimodal fake news detection methods can be broadly categorized into point-based representation and distribution-based representation. Point-based methods primarily focus on fusing and interacting multimodal features through strategies such

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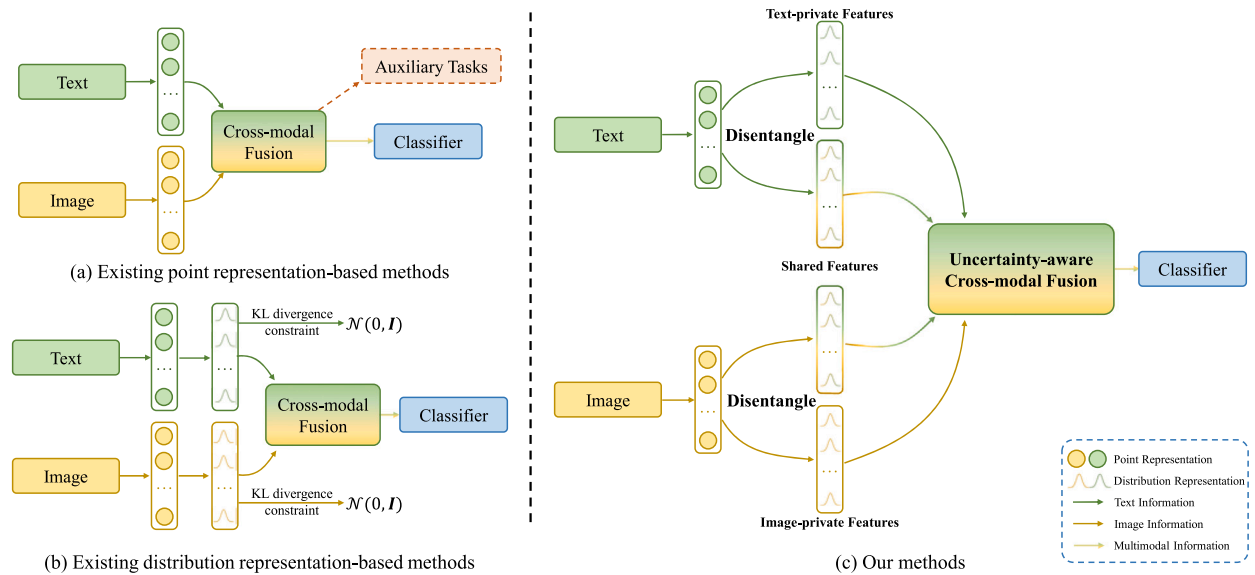


Fig. 1. Comparison between our method and existing methods.

as concatenation, element-wise operations, and attention mechanisms (Song, Ning, Zhang, & Wu, 2021; Wu, Long, Gao, Wang, & Zhang, 2023; Xue et al., 2021; Zhou, Wu, & Zafarani, 2020), while also introducing auxiliary tasks like multi-task learning and contrastive learning to capture inter-modal correlations (Chen et al., 2022; Peng, Jian, Li, & Shen, 2023; Zeng et al., 2023). On the other hand, distribution-based methods, exemplified by Khattar, Goud, Gupta, and Varma (2019) and Wei, Hu, Zhou, and Hu (2023), employ probabilistic frameworks such as Variational Autoencoders (VAEs) to model the distributions of multimodal features.

However, despite the promising results achieved by existing methods, they still face several challenges:

(1) **Inability to capture uncertainty of multimodal data:** Existing methods primarily rely on deterministic representations, which overlook the inherent ambiguity and vagueness in multimodal data, limiting their ability to capture nuanced relationships. This uncertainty exists both within individual modalities where words or images may carry multiple meanings depending on context and across modalities, where mismatched text-image pairs may create further ambiguity. For instance, an image may not perfectly align with its accompanying text, either unintentionally or as part of deceptive content, complicating accurate interpretation. Addressing both intra- and cross-modal uncertainties is essential for models to flexibly and reliably analyze complex multimodal data. Although some researchers have attempted to address this issue (Khattar et al., 2019; Wei et al., 2023) by using Variational Autoencoder-based models with MLPs to capture multimodal uncertainty, MLPs often lack the capability to effectively model complex, fine-grained interactions across modalities. Moreover, they enforce the latent representations to approximate a standard Gaussian distribution, which can oversimplify the latent space and limit its capacity to model the complex structures inherent in multimodal data, leaving this limitation an area that remains further explored.

(2) **Heterogeneity across modalities:** Information from different modalities in multimodal fake news detection, such as text and images, introduces significant challenges due to redundancy and distribution gaps. Information from each modality may overlap, such as an image reinforcing details already present in the text, leading to redundant data that can burden the model and complicate feature integration (Ghorbanali & Sohrabi, 2024). Additionally, the inherent differences in how each modality represents information – such as linguistic for text and visual for images – create distribution gaps that make it difficult to align data effectively (Radford et al., 2021). These gaps can result in misaligned representations, reducing the accuracy and robustness of fake news detection models. Addressing this heterogeneity is essential for improving multimodal integration, reducing redundancy, and ensuring more reliable detection.

(3) **Imbalance between attention to modality-private veracity and cross-modal inconsistencies:** In multimodal fake news detection, there is often an imbalance in the attention given to the veracity of content within each modality versus the inconsistencies across modalities. Many methods emphasize detecting mismatches between text and images as signals of deception (Sun, Zhang, Ma, et al., 2023; Xue et al., 2021), but this can come at the expense of thoroughly assessing the authenticity of each modality independently. Such an approach risks missing key indicators of fake news embedded within individual modalities, as it prioritizes cross-modal alignment over in-depth analysis of each modality's inherent truthfulness. A more balanced focus that considers both the internal veracity of each modality and potential cross-modal inconsistencies could enhance detection accuracy and robustness.

To address the above limitations, we propose a novel Uncertainty-aware Disentangled Representation Learning (UDRL) framework for multimodal fake news detection. First, to capture the uncertainty and ambiguity in multimodal data, we propose a probabilistic representation module (PRM) that models features using Gaussian distributions instead of deterministic points, enabling the model to handle intra-modal uncertainty and ambiguity. Second, to address modality heterogeneity, we adopt a disentangled

representation learning framework that separates shared and private information for more robust and discriminative representations. A shared encoder extracts cross-modal representations, guided by an inter-class triplet loss to separate different classes and an intra-class center loss to cluster representations of the same class. Private encoders capture unique modality-specific features, while an information redundancy constraint minimizes mutual information between shared and private representations to reduce redundancy. Additionally, image and text decoders reconstruct the input data, ensuring essential information is preserved. This framework bridges the distribution gap across modalities and ensures effective multimodal integration. At the same time, the separation of shared and private information balances attention between modality-private veracity and cross-modal inconsistencies, enabling a comprehensive evaluation of each modality and their interactions. Finally, we propose an uncertainty-aware fusion module that uses learned weighting factors based on the uncertainty of the modality representations to dynamically adjust the importance of each modality during the fusion process, achieving better cross-modal interaction.

Our contributions can be summarized as follows:

- (1) We propose a probabilistic representation module to model the representations of multimodal information as Gaussian distributions, which can capture the uncertainty and ambiguity of multimodal information.
- (2) We introduce a disentangled representation learning framework to disentangle the shared and private information across modalities, which can learn more robust and discriminative representations.
- (3) We propose an uncertainty-aware fusion module that leverages the uncertainty of the multimodal representations to guide the fusion process, which can achieve better cross-modal interaction according to the uncertainty of the modality representations.
- (4) Extensive experiments on three benchmark datasets demonstrate that our method achieves consistently strong performance, validating its effectiveness in multimodal fake news detection.

2. Related works

2.1. Fake news detection

Early fake news detection methods mainly rely on either text or image content (Ahmed, Ali, Hussain, Baseer, & Ahmed, 2021; Han, Han, Zhang, Li, & Cao, 2021; Kapusta, Drlik, & Munk, 2021; Kaur, Kumar, & Kumaraguru, 2020; Liu, Lian, Gu, & Xiao, 2022). For example, Qi, Cao, Yang, Guo, and Li (2019) concentrate on the frequency and pixel domain features in images to detect fake news. Shrivastava, Singh, Jain, and Kaushal (2022) explore the use of bag of words and tf-idf vectorization to extract textual features for detection. However, the single modality analysis cannot make full use of multimodal information. Researchers have increasingly focused on multimodal fake news detection, which combines textual, visual, and social context to enhance detection performance (Khatter et al., 2019; Kumari & Ekbal, 2021; Qian, Wang, Hu, Fang, & Xu, 2021; Wang et al., 2018; Yang et al., 2022; Yu, Liu, Zhu, Li, & Luo, 2024). Initially, simple concatenation of image and text features is used (Singhal, Dhawan, Shah, & Kumaraguru, 2021; Wang et al., 2018), which cannot capture the complex relationships between different modalities. Recently, many researchers have proposed to use attention mechanism to fuse multimodal features (Jing, Wu, Sun, Fang, & Zhang, 2023; Li et al., 2024; Luvembe, Li, Li, Liu, & Wu, 2024; Paka, Bansal, Kaushik, Sengupta, & Chakraborty, 2021). For instance, Hu, Zhao, Qi, Song, and Nie (2024) put forward an image-text matching-aware co-attention mechanism which utilizes multimodal representations captured by ViLT (Kim, Son, & Kim, 2021). Yang, Zhang, Zhang, Cheng and Hu (2024) propose a relationship-aware attention network to fuse multi-level text features and image features for effective fake news detection. Luvembe et al. (2024) introduce a complementary attention through image captioning and a scaled dot-product mechanism to capture fine-grained cross-modal relationships. Bai, Liu, and Li (2024) propose a frequency-aware parallel cross-modal attention structure to capture interdependencies among frequency, spatial, and text domains.

Moreover, some researchers incorporate auxiliary tasks to aid detection (Hua, Cui, Li, Tang, & Zhu, 2023; Peng et al., 2023; Wang, Li, Li, Wang and Gao, 2024). For instance, Yang, Bao et al. (2023) employ Wasserstein distance to capture intermodal discrepancies between visual and textual embeddings, enhancing rumor detection through combined modal fusion and inconsistency measurement for a more comprehensive prediction. Wang, Li et al. (2024) introduce a unimodal veracity classification task, framed as a positive and unlabeled learning problem, to assign veracity labels to each modality, enhancing the discriminative power of multimodal features for fake news detection. Sengan et al. (2023) incorporate a CNN-based knowledge extractor to extract stance from implicitly related post-reply pairs and integrate these stance representations with other features to enhance fake news detection accuracy. As event detection has proven effective in identifying key structural patterns in dynamically evolving information streams (Lin, Xu, Yu, Tian, & Luo, 2023), recent studies have explored its potential in fake news detection. Ma et al. (2024) propose Event-Radar, which models subject-predicate relationships via an event graph and integrates multimodal cues to improve detection robustness in noisy news environments. Additionally, contrastive learning has been explored to enhance multimodal fake news detection. Shao, Wu, Wu, Nie, and Liu (2024) propose HAMMER, which employs manipulation-aware contrastive learning to improve cross-modal alignment and enhance the model's ability to capture subtle inconsistencies.

Recent advancements in large language models (LLMs) have shown their potential in various tasks, including fake news detection, due to their rich knowledge base, interpretability, and reasoning abilities. Specifically, Xie, Zhu, Han, and Hu (2024) utilized LLMs for knowledge retrieval, evaluating whether retrieved information supports fake news classification. Yang, Zhang, Gao, and Zhang (2024) enhanced fake news detection by generating diverse semantic variations and adversarial perturbations using LLMs. Additionally, MMIDR (Wang, Xu, Zhang et al., 2024) applies multimodal LLMs for image description and information retrieval to improve multimodal fake news detection. Jiang and Wang (2024) enhanced LLM reasoning for fake news detection by incorporating context learning to improve decision-making accuracy.

However, existing methods suffer from overlooking the inconsistent information from different modalities in news, often focusing excessively on addressing these inconsistencies while neglecting the inherent truthfulness of the content itself. Recently, some researchers have considered these two challenges. Zhou, Yang, Ying, Qian, and Zhang (2023) utilize ResNet-101 (He, Zhang, Ren, & Sun, 2016) and BERT (Devlin, Chang, Lee, & Toutanova, 2018) to capture single-modality features and employ the CLIP model (Radford et al., 2021) to capture cross-modal features. However, CLIP can only process up to 77 tokens, limiting its capacity to fully capture textual context. Additionally, the multimodal information can overlap with single-modality features, leading to redundant data and unnecessary information loss. To address this issue, this paper approaches the problem from the perspective of disentangled learning, aiming to minimize information overlap while examining news from multiple views.

2.2. Probabilistic embedding

Probabilistic embedding in deep learning is characterized by the use of probabilistic methods to represent and process uncertainty within deep learning frameworks. The core idea of these methods is to utilize probabilistic distribution instead of point estimation to represent input in feature space, which has been validated to improve interpretability and robustness in various tasks including face recognition (Chang, Lan, Cheng, & Wei, 2020), speaker diarization (Yang, Chen, Wang and Wang, 2023), human pose embedding (Sun et al., 2020), and person re-identification (Yu, Li, Yang, Hospedales, & Xiang, 2019). In the context of multimodal tasks, probabilistic embeddings play a crucial role in effectively managing and integrating diverse data types by capturing and representing uncertainties inherent in multimodal data. For example, the Probvlm (Upadhyay, Karthik, Mancini, & Akata, 2023) framework employs probabilistic adapters for frozen vision-language models, enhancing their adaptability and robustness by leveraging uncertainty information. In cross-modal retrieval, probabilistic embedding improves the alignment and retrieval process by representing data across different modalities using probabilistic distributions (Chun, Oh, De Rezende, Kalantidis, & Larlus, 2021). The MAP framework (Ji et al., 2023) exemplifies the application of probabilistic embeddings in vision-language pre-training, where uncertainty-aware mechanisms enhance model robustness. COLD model (Tellamekala et al., 2023) integrates audio and visual data for emotion recognition using calibrated and ordinal latent distributions to account for variability in emotional expressions.

As for multimodal fake news detection, Wei et al. (2023) utilize a variational autoencoder-based encoder to model multimodal content, capturing uncertainty through probabilistic latent representations. However, one limitation of their approach is that it primarily focuses on modeling the uncertainty at the level of global latent variables, while neglecting the important context-specific interactions between individual tokens within each modality. This lack of contextual modeling between tokens limits the ability to capture fine-grained uncertainty at the token level, which is crucial for understanding complex, nuanced relationships within the multimodal data and improving detection accuracy.

2.3. Disentangle representation learning

The core idea of Disentangle Representation learning is to separate the underlying factors of variation in data into distinct, interpretable variables, enhancing model explainability and generalization, which has been validated in various domains including computer vision (Dalva, Pehlivan, Hatipoglu, Moran, & Dundar, 2023; Liang et al., 2023), natural language processing (Miao, Zhou, Pavlovski, & Qian, 2024; Song, Giunchiglia, Li, Tian, & Xu, 2024), and multimodal tasks (Huang et al., 2024; Zhang, Yang, Zhou, & Wang, 2024). In computer vision, RevColV2 (Han, Cai, & Zhang, 2023) disentangles low-level and semantic features within its autoencoder architecture, allowing for improved representation consistency during masked image modeling and enhancing performance across various downstream vision tasks. As for NLP, VDR (Zhou et al., 2024) decouples intrinsic data characteristics from their linguistic counterparts, enhancing performance in retrieval tasks by leveraging natural language to facilitate effective disentangled representation learning. In multimodal tasks, disentangled learning is used to better manage intrinsic heterogeneity across modalities (Wang, Xu, Wei et al., 2024; Zeng, Yan, Mai, & Hu, 2024), while enhancing feature integration and reducing information redundancy (Li et al., 2023; Sun et al., 2024).

In the context of fake news detection, Wang et al. (2023) introduce a novel approach that focuses on disentangling fake news propagation on social media to infer the underlying intentions of responsive posts. However, the application of disentangled learning in content-based multimodal fake news detection remains unexplored. In this study, we propose to extend these concepts by integrating disentangled representation learning to effectively manage and utilize multimodal content, thereby improving the detection performance of fake news through better understanding of the interactions between different modalities.

3. Methodology

Our proposed Uncertainty-aware Disentangled Representation Learning (UDRL) framework consists of three main components, as shown in Fig. 2. Firstly, the input image V and text T are fed into pretrained Swin Transformer (Liu et al., 2021) and BERT to obtain the image and text representations X_V and X_T , respectively (Section 3.1). Then, we introduce a probabilistic representation encoder module to model the representations of multimodal information as Gaussian distributions, which can capture the uncertainty and ambiguity of multimodal data. (Section 3.2) Next, the image and text representations are fed into the modality-shared probabilistic representation encoder E_S and modality-private probabilistic representation encoders E_V and E_T to obtain the shared probabilistic representations z_s^u, z_s^l and private probabilistic representations z_p^u, z_p^l for image and text, respectively. To ensure proper disentanglement, we introduce three constraints: the multimodal consistency constraint to align shared representations across modalities, the information redundancy constraint to minimize overlap between shared and private representations, and the

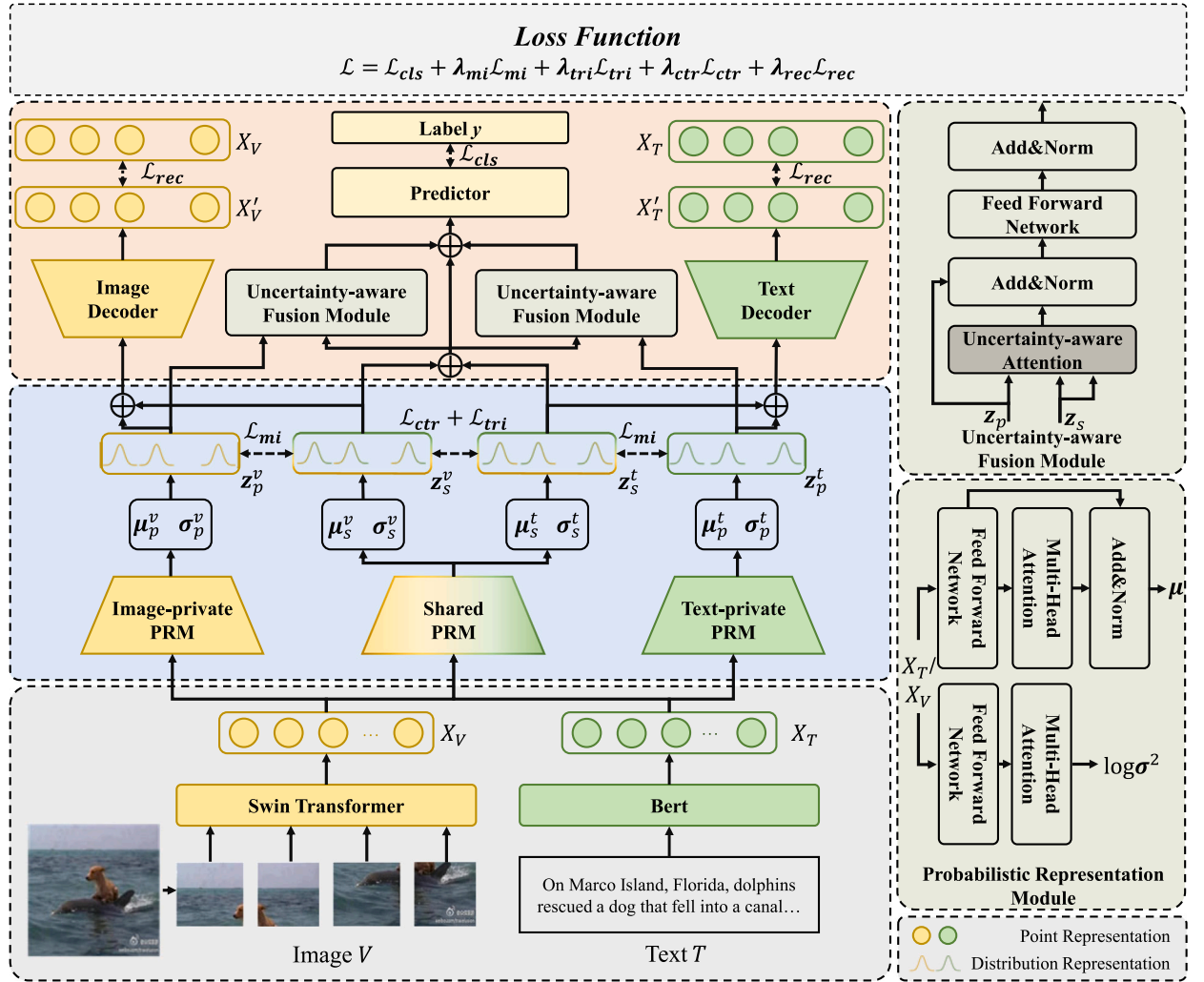


Fig. 2. The architecture of the proposed Uncertainty-aware Disentangled Representation Learning (UDRL) framework. The framework consists of four main components: (1) Unimodal Feature Extraction (gray block): Image input V and text input T are processed by Swin Transformer and BERT respectively to obtain initial representations X_V and X_T ; (2) Probabilistic Representation Module (including in blue block): Transforms inputs into Gaussian distributions to capture data uncertainty; (3) Disentangled Representation Learning (including in blue block): Both modality-shared encoder E_S and modality-private encoders E_V and E_T disentangle shared information (z_p^v, z_p^t) from modality-private information (z_s^v, z_s^t) while enforcing multimodal consistency, minimizing information redundancy, and ensuring accurate reconstruction; (4) Uncertainty-aware Fusion Module and Predictor (light orange block): Leverages uncertainty estimates to guide the cross-modal fusion process, adaptively weighting modality contributions based on their reliability for final prediction.

multimodal reconstruction constraint to ensure accurate modality reconstruction (Section 3.3). Finally, we propose an uncertainty-aware fusion module to leverage the uncertainty of the multimodal representations to guide the fusion process, which can achieve better cross-modal interaction according to the uncertainty of the modality representations (Section 3.4). The final prediction is made based on the fused representations and is optimized by the combined loss of the task loss and the disentangled representation learning loss (Section 3.6).

3.1. Unimodal feature extraction

Given the input image V and text T , we first extract the image and text representations X_V and X_T using pretrained transformer-based models. For image V , we adopt the Swin Transformer pretrained on large-scale image datasets to extract the image representation and transform it to sequence embeddings $X_V \in \mathbb{R}^{n \times d_v}$:

$$X_V = \text{SwinT}(V) = \{x_1^v, x_2^v, \dots, x_n^v\} \quad (1)$$

where n is the number of image patches and $x_i^v \in \mathbb{R}^{d_v}$ is the representation of the i th image patch, d_v is the hidden dimension of the image representation.

For text T , we adopt the BERT pretrained on large-scale text corpora to extract the text representation $X_T \in \mathbb{R}^{m \times d_t}$. The text representation X_T is obtained by feeding the text T into the BERT model:

$$X_T = \text{BERT}(T) = \{x_1^t, x_2^t, \dots, x_m^t\} \quad (2)$$

where m is the number of tokens in the text and $x_i^t \in \mathbb{R}^{d_t}$ is the representation of the i th token, d_t is the hidden dimension of the text representation.

3.2. Probabilistic representation module

To represent the uncertainty and ambiguity of multimodal information, in this section we propose a probabilistic representation module to model the representations of multimodal information as Gaussian distributions. There are two main reasons for using Gaussian distributions to model the representations of multimodal information. First, Gaussian distributions offer elegant mathematical properties, making them easy to work with and interpret in modeling and analysis. Second, Gaussian distributions are widely applicable across various domains due to their common occurrence in natural phenomena, simplifying uncertainty modeling. Further discussion on the choice of distribution is provided in the Experiments section (Section 4.6.2).

The architecture of proposed probabilistic representation module is shown in Fig. 2. Inspired by the transformer block, we leverage the multi-head attention mechanism and feed-forward neural networks to obtain the mean and variance of the representation distribution. For the mean computation, we employ layer normalization and residual connections after the attention mechanism, which preserves the original feature information. In contrast, for the variance computation, we utilize a similar structure but intentionally omit the residual connection, allowing the network to focus specifically on modeling the uncertainty in the representations. This approach effectively captures both the central tendency and variability in the feature representations, enhancing the robustness and performance of our model. Below, we describe the process of obtaining the mean and variance using the image representation as an example.

To calculate the mean of the Gaussian distribution, we first apply a feed-forward network to the input image representation X_V :

$$\hat{X}_V^\mu = \text{FeedForward}(X_V) \quad (3)$$

The Feed Forward layer is designed to facilitate interactions at the feature level, enriching the representation by capturing complex feature dependencies. This result is then passed through a multi-head attention layer and combined with the original input through a residual connection, followed by layer normalization to produce the mean μ :

$$\mu_V = \text{LN}(\hat{X}_V + \text{MultiHead}(\hat{X}_V^\mu)) \quad (4)$$

For the variance calculation, we follow a similar process. The input X_V is first passed through another feed-forward network to compute the variance-specific features:

$$\hat{X}_V^\sigma = \text{FeedForward}(X_V) \quad (5)$$

The Multi-Head operation then captures token-level dependencies, allowing the model to attend to multiple aspects of the input sequence simultaneously. These features are then processed by a multi-head attention layer, added to the original features via a residual connection, and normalized to produce $\log \sigma^2$, which represents the log variance:

$$\log \sigma_V^2 = \text{MultiHead}(\hat{X}_V^\sigma) \quad (6)$$

where $\mu_V \in \mathbb{R}^{n \times d_v}$ and $\sigma_V \in \mathbb{R}^{n \times d_v}$ are the mean and variance of the Gaussian distribution of image representation, respectively. The mean and variance of the Gaussian distribution for text representation can be obtained in a similar manner, denoted as μ_T and σ_T , respectively.

3.3. Disentangled representation learning

3.3.1. Modality-shared and -private representation learning

While existing fake news detection methods straightly fuse the features extracted through pretrained BERT, ResNet, CLIP, etc., we argue that the heterogeneity between modalities results in the less effectiveness in fusing multimodal features. Inspired by these observations, we propose a disentangled representation learning framework to disentangle the shared and private information across modalities, enabling the model to learn more robust and discriminative representations.

We denote the shared encoder as E_S and the private encoders for image and text as E_V and E_T , respectively. The shared encoder E_S is used to learn the modality-shared representations, which can capture the shared information across modalities. The private encoders E_V and E_T are used to learn the modality-private representations, which can capture the modality-private information. Both the shared and private encoder are implemented as proposed probabilistic representation module in Section 3.2.

The mean and variance of the shared probabilistic representations z_s^v, z_s^t and private probabilistic representations z_p^v, z_p^t for image and text can be obtained as follows:

$$\begin{aligned} \mu_m^n &= E_m^n(X_m) \\ \sigma_m^n &= E_m^n(X_m) \end{aligned} \quad (7)$$

where $m \in \{s, p\}$ denotes the shared encoder E_S and private encoders E_V and E_T , respectively. $n \in \{v, t\}$ denotes the image and text modality, respectively.

Therefore, we can obtain the distributions of the shared and private representations as follows:

$$\begin{aligned} p(z_s^v | X^V) &\sim \mathcal{N}(\mu_s^v, \text{diag}(\sigma_s^v)) \\ p(z_s^t | X^T) &\sim \mathcal{N}(\mu_s^t, \text{diag}(\sigma_s^t)) \\ p(z_p^v | X^V) &\sim \mathcal{N}(\mu_p^v, \text{diag}(\sigma_p^v)) \\ p(z_p^t | X^T) &\sim \mathcal{N}(\mu_p^t, \text{diag}(\sigma_p^t)) \end{aligned} \quad (8)$$

where $\text{diag}(\sigma)$ refers to a diagonal matrix where the diagonal entries are the elements of the vector σ .

3.3.2. Information redundancy constraint

To ensure that the shared and private representations are disentangled with minimal redundancy, we introduce an independence constraint to reduce mutual information, thereby minimizing redundancy between these two representations. Due to its ability to capture the geometric discrepancy between distributions, we employ the Wasserstein distance between the Gaussian distributions of the shared and private representations as an effective measure of dependency, encouraging minimal overlap and redundancy. Taking the text modality as an example, for two Gaussian distributions $p(z_s^t | X^T)$ and $p(z_p^t | X^T)$, the closed-form expression of the 2-Wasserstein distance is given by:

$$D_{W_2}(p(z_s^t | X^T), p(z_p^t | X^T)) = \|\mu_p^t - \mu_s^t\|^2 + \text{Tr}(\sigma_p^t + \sigma_s^t - 2(\sigma_p^{t/2} \sigma_s^{t/2})^{1/2})^{1/2} \quad (9)$$

Our goal is to reduce the redundancy between the shared and private representations, which can be achieved by maximizing this approximation of mutual information. To enforce this, we introduce a information redundancy constraint loss function defined as:

$$\mathcal{L}_{mi} = e^{-D_{W_2}(z_s^v, z_p^v)} + e^{-D_{W_2}(z_s^t, z_p^t)} \quad (10)$$

This constraint encourages the shared and private representations to be independent by reducing redundancy, thereby improving the disentanglement and overall performance of our model.

3.3.3. Consistency constraint

To ensure alignment of the shared representations across different modalities, we introduce a hybrid loss combining inter-class triplet loss and intra-class center loss. The inter-class triplet loss brings representations of the same class closer while pushing apart those from different classes, focusing on pairwise relationships. In contrast, the intra-class center loss minimizes intra-class variance by pulling representations toward the class center, ensuring compactness within each class. Together, these losses enhance both class separation and consistency.

Specifically, the inter-class triplet loss is formulated as follows:

$$\mathcal{L}_{tri} = \sum_{(a,p,n) \in \mathcal{T}} \left[\max(0, \alpha + D_{KL}(z_{sa}^v, z_{sn}^t) - D_{KL}(z_{sa}^v, z_{sp}^t)) + \max(0, \alpha + D_{KL}(z_{sa}^t, z_{sn}^v) - D_{KL}(z_{sa}^t, z_{sp}^v)) \right] \quad (11)$$

where z_{sa}^v and z_{sa}^t represent the shared representations of the anchor sample in the visual and text modalities, respectively; z_{sp}^v and z_{sp}^t denote the positive samples from the same class; z_{sn}^v and z_{sn}^t denote the negative samples from a different class; α is the margin that encourages separation between inter-class representations; and $D_{KL}(\cdot \| \cdot)$ represents the KL divergence between two Gaussian distributions $\mathcal{N}(\mu_1, \Sigma_1)$ and $\mathcal{N}(\mu_2, \Sigma_2)$, computed as:

$$D_{KL}(\mathcal{N}(\mu_1, \Sigma_1) \| \mathcal{N}(\mu_2, \Sigma_2)) = \frac{1}{2} \left(\text{Tr}(\Sigma_2^{-1} \Sigma_1) + (\mu_2 - \mu_1)^T \Sigma_2^{-1} (\mu_2 - \mu_1) - d + \log \frac{\det \Sigma_2}{\det \Sigma_1} \right) \quad (12)$$

where d is the dimensionality of the Gaussian distributions and $\text{Tr}(\Sigma_2^{-1} \Sigma_1)$ is the trace of the matrix product $\Sigma_2^{-1} \Sigma_1$, measuring the similarity in shape between the covariance matrices Σ_1 and Σ_2 ; and $\det$ represents the determinant of the covariance matrix.

However, the inter-class triplet loss alone only considers the separation between classes and may not be sufficient to ensure robust alignment within each class. Therefore, inspired by prior work, we introduce an intra-class center loss to further minimize intra-class variance in the shared representations. The intra-class center loss is defined as:

$$\mathcal{L}_{ctr} = \frac{1}{C} \sum_{i=1}^C \sqrt{\frac{\sum_{j=1}^N \tau_j (\mu_j - \mu_{\text{weighted}})^2}{\sum_{j=1}^N \tau_j}} \quad (13)$$

where C is the number of classes, N is the number of samples within each class, τ_j is the weight of the j th distribution given by $\tau_j = \frac{1}{\sigma_j^2}$, μ_j is the mean of the j th shared representation, and μ_{weighted} is the weighted mean of the class, calculated as $\mu_{\text{weighted}} = \frac{\sum_{j=1}^N \tau_j \mu_j}{\sum_{j=1}^N \tau_j}$.

Finally, the consistency constraint combines the inter-class triplet loss and the intra-class center loss, and is expressed as:

$$\mathcal{L}_{consistency} = \mathcal{L}_{tri} + \mathcal{L}_{ctr} \quad (14)$$

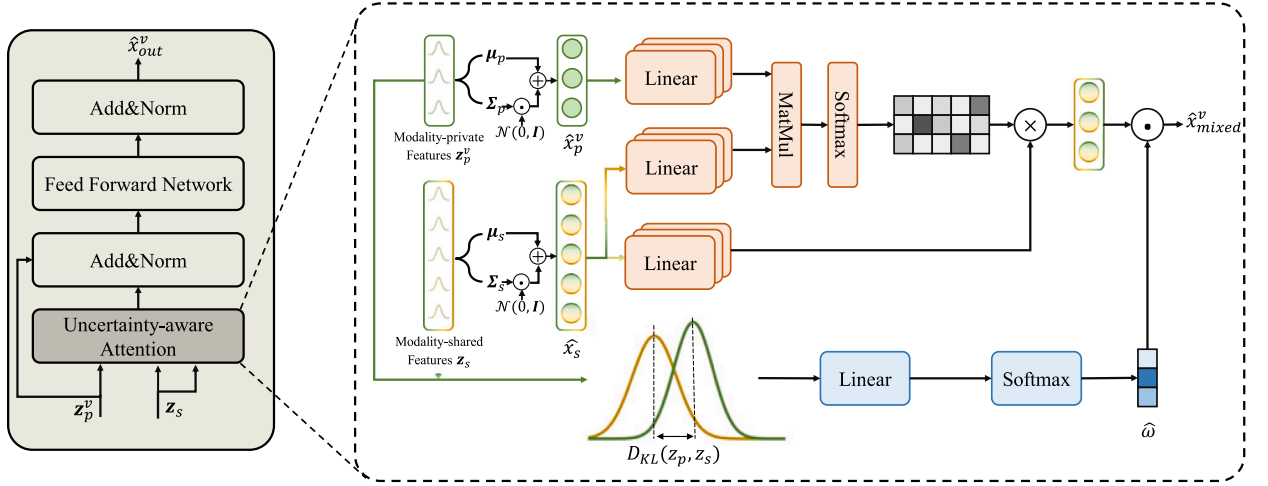


Fig. 3. The architecture of the proposed Uncertainty-aware Fusion Module. The framework consists of two main components: (1) Uncertainty-aware Attention: Implements a cross-attention mechanism where shared representations ($\hat{x}_s$) guide private representations ($\hat{x}_p^v$ or $\hat{x}_p^t$), with KL divergence between private and shared Gaussian distributions serving as a dynamic weighting factor—adjusting each modality's contribution based on uncertainty and information uniqueness; (2) A series of residual connections, Layer Normalization (LN), and a Feed-Forward Network (FFN): Refines the mixed representations through residual connections that preserve essential information, Layer Normalization that stabilizes training by reducing internal covariate shift, and a Feed-Forward Network that captures complex interactions through non-linear transformations, collectively yielding robust outputs that effectively incorporate cross-modal uncertainty.

3.3.4. Information reconstruction constraint

To distinguish the shared and private representations and reduce information loss during the disentangled representation learning process, we introduce the information reconstruction constraint. Specifically, we leverage image and text decoder to reconstruct the input representation, which can further ensure the information is well preserved during the disentangled representation learning process. The reconstruction loss can be formulated as:

$$\mathcal{L}_{rec} = \mathbb{E}_{z_s^v, z_s^t, z_p^v, z_p^t} [\|X_V - \text{Decoder}_V(z_s^v, z_p^v)\|_2^2 + \|X_T - \text{Decoder}_T(z_s^t, z_p^t)\|_2^2] \quad (15)$$

where Decoder_V and Decoder_T are the image and text decoder, respectively. The image and text decoder are implemented as feed-forward neural networks, which can reconstruct the input image and text representation, respectively.

3.4. Uncertainty-aware multimodal fusion

Existing disentangled representation learning methods often concatenate shared and private representations to fuse multimodal information. However, simple concatenation may not sufficiently capture complex cross-modal relationships, especially when the uncertainty and ambiguity inherent in multimodal data are not considered. To address this limitation, we propose an uncertainty-aware fusion module that leverages the uncertainty of multimodal representations to guide the fusion process, achieving enhanced cross-modal interaction by adapting to the uncertainty associated with each modality.

As shown in Fig. 3, the proposed uncertainty-aware fusion module begins by generating modality-private private and shared features, denoted as z_p^v , z_p^t , and z_s , respectively. These features are modeled as Gaussian distributions to capture the uncertainty of each modality, represented by a mean μ and covariance Σ . To enable gradient-based optimization, we apply the reparameterization trick, introducing reparameterized vectors $\hat{x}_p^v \in \mathbb{R}^{n \times d_v}$, $\hat{x}_p^t \in \mathbb{R}^{m \times d_t}$, and $\hat{x}_s \in \mathbb{R}^{(m+n) \times d_s}$ as follows:

$$\hat{x}_p^v = \mu_p^v + \Sigma_p^{v/2} \cdot \epsilon_v, \quad \hat{x}_p^t = \mu_p^t + \Sigma_p^{t/2} \cdot \epsilon_t, \quad \hat{x}_s = \mu_s + \Sigma_s^{1/2} \cdot \epsilon_s \quad (16)$$

where $\epsilon_v, \epsilon_t, \epsilon_s \sim \mathcal{N}(0, I)$ are random variables sampled from standard normal distributions, $\mu_p^v, \Sigma_p^v, \mu_p^t, \Sigma_p^t, \mu_s$, and Σ_s are the means and covariances of the modality-private and modality-shared features.

Following reparameterization, we apply a cross-attention mechanism where the shared representation $\hat{x}_s$ guides the private representation $\hat{x}_p^v$ (or $\hat{x}_p^t$ in the case of text). Take image modality as example, we transform $\hat{x}_p^v$ and $\hat{x}_s$ into queries, keys, and values, then compute the attention weights for each head i as follows:

$$\begin{cases} Q_i = \text{Linear}_{Q_i}(\hat{x}_p), & K_i = \text{Linear}_{K_i}(\hat{x}_s), & V_i = \text{Linear}_{V_i}(\hat{x}_s) \\ \text{Attention}_i(\hat{x}_p, \hat{x}_s) = \text{Softmax}\left(\frac{Q_i K_i^T}{\sqrt{d_k}}\right) V_i \\ \text{MultiHead}(\hat{x}_p, \hat{x}_s) = \text{concat} [\text{Attention}_1, \dots, \text{Attention}_H] W \end{cases} \quad (17)$$

where Q_i , K_i , and V_i each have a dimensionality of d_k , representing the dimension of each attention head. H is the number of attention heads, allowing the model to capture diverse features. $W \in \mathbb{R}^{H \times d_k \times d_k}$ is a learned projection matrix that ensures the final

multi-head attention output has the desired dimensionality. Similarly, the fusion process for the private text features $\hat{x}_p^t$ follows the same procedure, where $\hat{x}_s$ guides the text-private features in a comparable manner.

To incorporate uncertainty into the fusion process, we use the Kullback–Leibler (KL) divergence between the Gaussian distributions of z_p and z_s as a weighting factor. While the reparameterization trick allows for differentiable sampling, it cannot fully capture the relative uncertainty between modalities. The KL divergence addresses this limitation by directly measuring the distributional discrepancy between the private and shared features, thus providing a nuanced weighting for the fusion. Take the image modality as an example, the KL divergence between z_p^v and z_s is computed as:

$$\begin{cases} D_{KL}(z_p^v \parallel z_s) = \frac{1}{2} \left(\text{Tr}(\Sigma_s^{-1} \Sigma_p^v) + (\mu_s - \mu_p^v)^T \Sigma_s^{-1} (\mu_s - \mu_p^v) - d + \log \frac{\det \Sigma_s}{\det \Sigma_p^v} \right) \\ \hat{w} = \text{Softmax}(\text{Linear}(D_{KL}(z_p^v \parallel z_s))) \end{cases} \quad (18)$$

where d is the dimensionality of the Gaussian distributions, μ_p^v and μ_s are the mean vectors, and Σ_p^v and Σ_s are the covariance matrices of the private visual features and the shared features, respectively. Here, $D_{KL}(z_p^v \parallel z_s)$ provides an uncertainty-based weight, which is further processed by a linear transformation followed by a Softmax activation to yield the refined weighting factor $\hat{w}$. This final weight $\hat{w}$ is then used to scale the fused representation, resulting in the mixed representation:

$$\hat{x}_{\text{mixed}}^v = \hat{w} \cdot \text{MultiHead}(\hat{x}_p^v, \hat{x}_s) \quad (19)$$

The KL divergence allows the model to adjust the weight of each modality's contribution based on how distinct the modality-private features are from the shared features. When the KL divergence is large, it indicates that the modality-private features contain unique information, and the model will place more emphasis on the modality-private features. On the other hand, a small KL divergence indicates that the modality-private features are well-aligned with the shared features, and the model can rely more on the shared information. This dynamic weighting mechanism helps the model handle modality heterogeneity effectively and adaptively adjust the contribution of each modality based on its uncertainty.

After obtaining the mixed representation, we refine it through a series of residual connections, Layer Normalization (LN), and a Feed-Forward Network (FFN), which can be expressed as:

$$\begin{cases} \hat{x}_{\text{res}}^v = \text{LN}(\hat{x}_{\text{mixed}}^v + \hat{x}_p^v), \\ \hat{x}_{\text{ffn}}^v = \text{FFN}(\hat{x}_{\text{res}}^v), \\ \hat{x}_{\text{out}}^v = \text{LN}(\hat{x}_{\text{res}}^v + \hat{x}_{\text{ffn}}^v) \end{cases} \quad (20)$$

Here, each residual connection combines the original input with the transformed representation, preserving essential information while integrating new features. Layer Normalization (LN), applied after each residual connection, stabilizes training by normalizing activations, thus reducing internal covariate shift. The Feed-Forward Network (FFN) enriches the representation through non-linear transformations, allowing the model to capture more complex interactions. This sequence of residual connections, LN, and FFN serves to normalize, enhance, and stabilize the fused representation, yielding a robust and informative output that incorporates cross-modal uncertainty. Similarly, the output for the text modality $\hat{x}_{\text{out}}^t$ is processed through the same sequence of steps.

3.5. Predictor

After obtaining the mixed representation of the multimodal features, we concatenate the private and shared representations from both the visual modality ($\hat{x}_{\text{out}}^v$) and the text modality ($\hat{x}_{\text{out}}^t$), as well as the shared representation across modalities ($\hat{x}_s$). This ensures that the global cross-modal consistency captured by $\hat{x}_s$ is retained alongside the modality-private features, enhancing the model's ability to utilize both shared and unique characteristics. The concatenated feature vector is then passed through a two-layer fully connected network to perform the final classification task. The network can be expressed as:

$$p = \text{Softmax}(\text{MLP}(\text{concat}(\hat{x}_{\text{out}}^v, \hat{x}_{\text{out}}^t, \hat{x}_s))) \quad (21)$$

where $\hat{x}_{\text{out}}^v$ and $\hat{x}_{\text{out}}^t$ are the final output representations of the visual and text modalities, respectively, after the fusion process, and $\hat{x}_s$ is the shared representation that captures cross-modal interactions.

The inclusion of $\hat{x}_s$ ensures that information learned from cross-modal interactions during the disentanglement process is directly incorporated into the prediction, complementing the modality-private features. The MLP consists of two layers, with the first layer applying a ReLU activation and the second layer producing the raw output score, which is passed through the Softmax function to yield the probability p , representing the likelihood that the input sample is fake news.

3.6. Task loss

The classification loss $\mathcal{L}_{cls}$ is a cross-entropy loss, which can be computed as:

$$\mathcal{L}_{cls} = \frac{1}{N} \sum_{i=1}^N p_i \cdot \log p_i + (1 - p_i) \cdot \log(1 - p_i) \quad (22)$$

where N refers to the number of samples, and p_i refers to the probability of the i th sample being fake news.

Table 1
Statistics of Weibo, Gossipcop and Politifact.

Datasets	Training set		Testing set		All
	Fake	Real	Fake	Real	
<i>Weibo</i>	3346	2807	863	835	7851
<i>Gossipcop</i>	3134	10,421	738	2651	16,944
<i>Politifact</i>	125	171	37	37	370

Therefore, the total loss of our model is as follows:

$$\mathcal{L} = \mathcal{L}_{cls} + \lambda_{mi} \cdot \mathcal{L}_{mi} + \lambda_{tri} \cdot \mathcal{L}_{triplet} + \lambda_{ctr} \cdot \mathcal{L}_{center} + \lambda_{rec} \cdot \mathcal{L}_{rec} \quad (23)$$

where λ_i refers to the coefficient of each loss.

4. Experiments

In this section, we detail the superiority of our proposed UDRL in multimodal fake news detection.

4.1. Datasets

In this study, we conduct experiments using three widely recognized benchmark datasets: Weibo, GossipCop, and PolitiFact. The Weibo dataset, collected from the popular Chinese microblogging platform, Sina Weibo, includes a mix of text, images, and videos, with both fake and real news labeled based on verification by Weibo's official fact-checking system and Xinhua News Agency from 2012 to 2016. The PolitiFact dataset consists of fact-checked articles sourced from PolitiFact, a leading platform that assesses the credibility of political claims in the U.S. This dataset spans from 2002 to 2018 and includes political statements and news reports. Similarly, GossipCop focuses on celebrity and entertainment news, aggregating articles related to U.S. celebrities from the GossipCop website, which specializes in detecting false information in the entertainment industry. The statistic of two datasets is shown in [Table 1](#).

4.2. Evaluation metrics

Adhere to existing research, we choose accuracy, precision, recall and f1 score to make fair and comprehensive comparison.

4.3. Baselines

We compare our UDRL with both traditional and recently published methods as competitive baselines:

- MVAE ([Khatter et al., 2019](#)) utilizes a Multimodal Variational Autoencoder to extract and combine textual and visual features for detecting fake news.
- SAFE ([Zhou et al., 2020](#)) examines the relationship between textual and visual features to identify fake news by detecting mismatched content in news articles.
- SpotFake+ ([Singhal et al., 2020](#)) leverages transfer learning to capture semantic and contextual information from both news articles and their associated images, offering a multimodal approach to detect fake news on full-length articles.
- CAFE ([Chen et al., 2022](#)) employs cross-modal alignment and ambiguity learning to extract and integrate multimodal features based on the level of cross-modal ambiguity.
- BMR ([Ying et al., 2023](#)) extracts representations from textual content, image patterns, and image semantics, using improved Multi-gate Mixture-of-Expert networks for feature refinement and cross-modal consistency prediction in fake news detection.
- MM-ULN ([Wei et al., 2023](#)) models both intra- and inter-modality uncertainty to enhance multimodal fake news detection, incorporating a variational attention fusion module that adapts to varying confidence levels across different modalities.
- MRML ([Peng et al., 2023](#)) uses deep metric learning to extract both intra-modal and inter-modal relationships for effective multimodal rumor detection by applying metric-based triplet learning within each modality and contrastive pairwise learning across modalities.
- MPFN ([Jing et al., 2023](#)) employs a progressive fusion network to capture and combine representational information from each modality at different levels, using a transformer-based visual feature extractor and a mixer to establish strong connections between modalities for multimodal disinformation detection.
- CAF-ODNN ([Luvembe et al., 2024](#)) captures subtle cross-modal relationships for multimodal fake news detection by introducing image captioning for semantic representation and using complementary attention fusion, with an optimized deep neural network to enhance feature extraction and preserve semantics across modalities.
- CSFND ([Peng, Jian, Kan, Qiao, & Li, 2024](#)) introduces contextual semantic representation learning by fusing local context features with global semantic features, effectively isolating semantically dissimilar fake news from real news in the detection process.

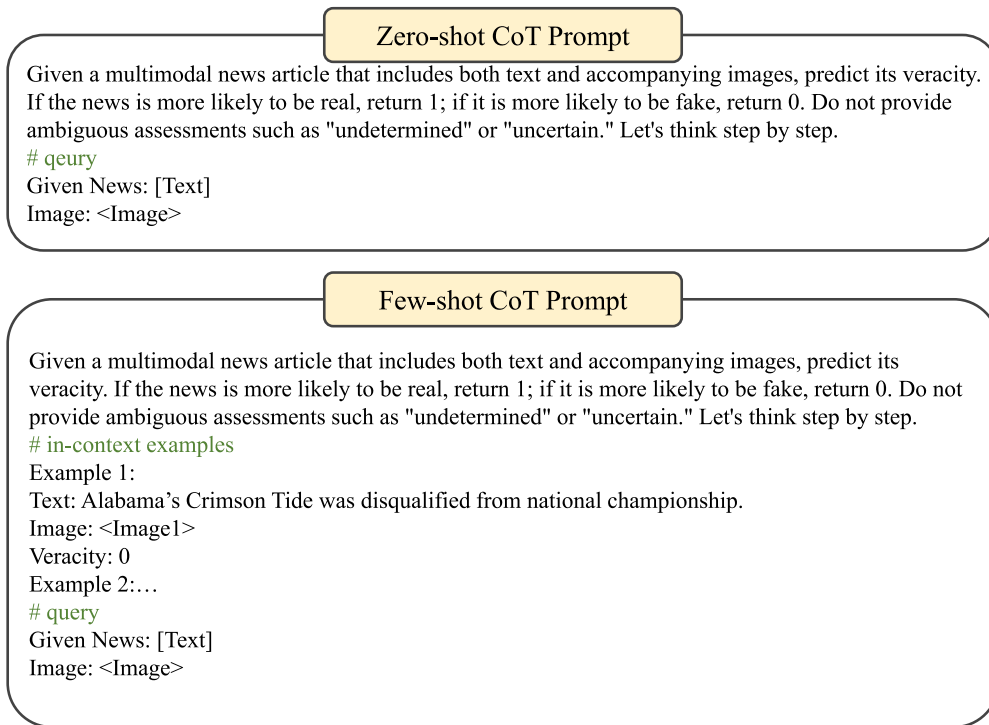


Fig. 4. Prompt used to ask GPT-4 to verify the multimodal news.

- FSRU (Lao et al., 2024) applies the Fourier transform to convert spatial features into the frequency domain, where it uses dual contrastive learning to obtain discriminative spectrum features for multimodal representation and fusion, enhancing rumor detection.
- MoPeD (Wang, Wu et al., 2024) proposes a model that addresses modality heterogeneity in multimodal fake news detection, which combines a dual encoding module, including a contrastive language-image pre-trained encoder, and a multilevel cross-modality fusion module to better handle the diverse features across text and images.
- Miner-uvs (Wang, Li et al., 2024) introduces a method that mitigates the neutralization effect in multimodal fake news detection by incorporating unimodal veracity signals, which predicts the veracity of each modality and uses these predictions to weigh the unimodal features, addressing conflicts between modalities during fusion.
- Event-Radar (Ma et al., 2024) presents an event-driven framework that integrates multi-view learning, focusing on event-level inconsistencies, which constructs an event graph to capture multimodal inconsistencies and uses a multi-view fusion mechanism to enhance robustness against poor-quality news.
- GPT-4 (Achiam et al., 2023) is a powerful large language model developed by OpenAI, known for its advanced capabilities in strong reasoning abilities and generating contextually relevant responses across a wide range of tasks.

4.4. Implementation details

Data preprocessing. To make a fair comparison, we follow the data preprocessing of Khattar et al. (2019). We removed noise data such as URLs, HTML entities, special punctuation marks, line breaks, and backslashes. We also converted all text to lowercase and replaced full-width characters with half-width characters, ensuring consistency and simplicity of the text, providing cleaner and more standardized input data for subsequent analysis.

Parameter Configuration. Following previous work to ensure a fair comparison, we divide each of the three datasets (Weibo, GossipCop, and PolitiFact) into training and testing subsets with an 8:2 ratio. For text content, each sentence is padded or truncated to have the same length. The token numbers are set to 162 for Weibo, 200 for GossipCop and PolitiFact. We use bert-base-chinese to extract features from Weibo and bert-base-uncased for GossipCop and PolitiFact. The dimension of token embeddings is 768. For image content, vit-base-ink221k is used to extract visual features. The token number of the image is 160, and the dimension of each token is 768.

In UDRL, the number of attention heads in the Probabilistic Representation Module and the uncertainty-aware fusion module is set to 8. The dropout rate is set to 0.1 to avoid overfitting. The training epoch is 30, with an early stop mechanism when the test accuracy does not improve within 5 epochs. Our model is trained using the AdamW optimizer, with a batch size of 32 and a

Table 2

Comparison between UDRL and different methods on Weibo, Gossipcop and Politifact datasets. The best performance is highlighted in **bold** and the second best is underlined.

Datasets	Models	Year	Accuracy	Fake news			Real news		
				Precision	Recall	F1	Precision	Recall	F1
Weibo	MVAE ^a	2019	0.824	0.854	0.769	0.809	0.802	0.879	0.870
	SAFE ^a	2020	0.816	0.818	0.815	0.817	0.816	0.818	0.817
	Spotfake+ ^b	2020	0.870	0.887	0.849	0.868	0.855	0.892	0.873
	CAFE ^c	2022	0.840	0.855	0.830	0.842	0.825	0.851	0.837
	BMR ^b	2023	0.914	0.932	0.895	0.913	0.896	0.933	<u>0.914</u>
	MRML ^b	2023	0.897	0.898	0.887	0.892	0.896	0.905	0.901
	MM-ULN ^b	2023	0.870	0.887	0.849	0.868	0.855	0.892	0.873
	MPFN ^b	2023	0.838	0.857	0.894	0.889	0.873	0.863	0.876
	CSFND ^b	2024	0.895	0.899	0.895	0.897	0.892	0.896	0.894
	FSRU ^b	2024	0.901	0.922	0.892	0.906	0.879	0.913	0.895
	MoPeD ^b	2024	0.883	0.879	0.887	0.883	0.887	0.878	0.883
	Miner-uvs ^a	2024	0.914	0.923	<u>0.906</u>	<u>0.915</u>	<u>0.905</u>	<u>0.922</u>	0.913
	Event-Radar ^b	2024	<u>0.919</u>	–	–	–	–	–	–
	GPT-4o (Zero-shot)	2024	0.657	0.649	0.657	0.653	0.665	0.657	0.661
	GPT-4o (Few-shot)	2024	0.674	0.666	0.675	0.671	0.682	0.673	0.678
	UDRL	–	0.926	<u>0.924</u>	0.931	0.927	0.928	0.921	0.924
Gossipcop	SAFE ^a	2020	0.838	0.758	0.558	0.643	0.857	0.937	0.895
	Spotfake+ ^b	2020	0.858	–	–	–	–	–	–
	CAFE ^c	2022	0.867	0.732	0.490	0.587	0.887	0.957	0.921
	BMR ^a	2023	0.881	<u>0.848</u>	0.554	0.670	0.886	0.972	0.927
	MRML ^a	2023	0.831	0.709	0.377	0.492	0.847	0.957	0.898
	CSFND ^a	2024	0.847	0.831	0.374	0.516	0.849	0.979	0.909
	FSRU ^a	2024	0.849	0.746	0.505	0.601	0.866	0.948	0.905
	Miner-uvs ^a	2024	<u>0.884</u>	0.815	<u>0.603</u>	<u>0.693</u>	<u>0.897</u>	0.962	<u>0.928</u>
	GPT-4o (Zero-shot)	2024	0.723	0.383	0.447	0.413	0.839	0.800	0.819
	GPT-4o (Few-shot)	2024	0.732	0.400	0.465	0.430	0.844	0.806	0.825
	UDRL	–	0.895	0.865	0.614	0.718	0.901	<u>0.973</u>	0.935
Politifact	SAFE ^c	2020	0.874	–	–	–	0.889	0.903	0.896
	Spotfake+ ^b	2020	0.856	–	–	–	–	–	–
	CAFE ^c	2022	0.838	0.758	0.558	0.643	0.857	0.937	0.895
	BMR ^a	2023	<u>0.905</u>	0.895	0.919	<u>0.907</u>	<u>0.917</u>	0.892	0.904
	MRML ^a	2023	0.878	0.868	0.892	0.880	0.889	0.865	0.878
	CSFND ^a	2024	0.851	0.882	0.811	0.845	0.825	0.892	0.857
	FSRU ^a	2024	0.839	0.846	0.866	0.853	0.829	0.804	0.811
	Miner-uvs ^a	2024	0.905	<u>0.917</u>	0.892	0.904	0.895	<u>0.919</u>	0.907
	GPT-4o (Zero-shot)	2024	0.747	0.765	0.722	0.743	0.730	<u>0.771</u>	<u>0.750</u>
	GPT-4o (Few-shot)	2024	0.775	0.794	0.750	0.771	0.757	0.800	0.778
	UDRL	–	0.919	0.919	0.919	0.919	0.919	<u>0.919</u>	0.919

^a Refers to the results from the re-implemented version of the original paper with open source code.

^b Refers to the results from the original paper.

^c Refers to the results from Zhou et al. (2023). – indicates that the results are not available.

learning rate of $2e-5$. The optimal hyperparameters of weighting coefficients in the loss function are determined by grid searching on the test set. For Weibo, GossipCop, and PolitiFact datasets, we set $\lambda_{mi} = 0.1$, $\lambda_{tri} = 0.1$, $\lambda_{ctr} = 0.1$, and $\lambda_{rec} = 0.1$. The margin is set to 0.5. All experiments are executed on a single NVIDIA GeForce RTX 4090D.

Considering the inability to access GPT-4's internal parameters for fine-tuning, we rely on prompt-based methods to evaluate its performance. Specifically, we set up zero-shot and few-shot settings for multimodal fake news detection. The prompt used is shown in Fig. 4.

4.5. Results

Table 2 displays the main experiment results of proposed UDRL and other competitive multimodal fake news detection methods. By conducting an in-depth analysis of these metrics, including accuracy, precision, recall, and F1 scores, we can derive several key observations:

(1) Improved Performance Over Traditional Methods: Our model demonstrates significant improvements over traditional methods such as MVAE, SAFE, and SpotFake+. This enhancement stems from the sophisticated design of our modules, which capture complex intra-modal and inter-modal interactions, rather than merely concatenating features across modalities. As a result, the model achieves stronger performance across various metrics.

(2) Effectiveness Compared to Competitive Methods: Our model demonstrates consistently strong performance when compared with recent competitive models such as BMR, CAF-ODNN, CSFND, FSRU, MoPeD, Miner-uvs, and Event-Radar. Specifically, compared to Miner-uvs, our model achieves consistent improvements across all datasets. Specifically, on the Weibo dataset, our

Table 3

Ablation study of different modules in UDRL on Weibo, Gossipcop and Politifact datasets.

Datasets	Models	Accuracy	Fake news		Real news	
			Precision	F1	Precision	F1
<i>Weibo</i>	w/o T	0.769 (+15.7%)	0.759 (+14.4%)	0.769 (+15.8%)	0.778 (+17.0%)	0.768 (+15.6%)
	w/o V	0.839 (+8.7%)	0.905 (+1.9%)	0.904 (+2.3%)	0.899 (+2.6%)	0.900 (+2.4%)
	w/o PRM	0.902 (+2.4%)	0.915 (+0.9%)	0.902 (+2.5%)	0.889 (+3.9%)	0.902 (+2.2%)
	w/o DRL	0.889 (+3.7%)	0.916 (+0.8%)	0.888 (+3.9%)	0.865 (+6.3%)	0.891 (+3.3%)
	w/o UAF	0.898 (+2.8%)	0.903 (+2.1%)	0.899 (+2.8%)	0.893 (+3.5%)	0.897 (+2.3%)
	UDRL	0.926	0.924	0.927	0.928	0.924
<i>Gossipcop</i>	w/o T	0.811 (+8.4%)	0.650 (+21.5%)	0.402 (+31.6%)	0.829 (+7.2%)	0.888 (+4.7%)
	w/o V	0.842 (+5.3%)	0.611 (+25.4%)	0.678 (+4.0%)	0.865 (+3.6%)	0.896 (+3.9%)
	w/o PRM	0.881 (+1.4%)	0.797 (+6.8%)	0.689 (+2.9%)	0.897 (+0.4%)	0.926 (+0.9%)
	w/o DRL	0.876 (+1.9%)	0.758 (+10.7%)	0.689 (+2.9%)	0.902 (−0.1%)	0.922 (+1.3%)
	w/o UAF	0.880 (+1.5%)	0.777 (+8.8%)	0.697 (+2.1%)	0.902 (−0.1%)	0.925 (+1.0%)
	UDRL	0.895	0.865	0.718	0.901	0.935
<i>Politifact</i>	w/o T	0.811 (+10.8%)	0.811 (+10.8%)	0.811 (+10.8%)	0.811 (+10.8%)	0.811 (+10.8%)
	w/o V	0.838 (+8.1%)	0.791 (+12.8%)	0.850 (+6.9%)	0.903 (+1.6%)	0.824 (+9.5%)
	w/o PRM	0.865 (+5.4%)	0.865 (+5.4%)	0.865 (+5.4%)	0.865 (+5.4%)	0.865 (+5.4%)
	w/o DRL	0.851 (+6.8%)	0.861 (+5.8%)	0.849 (+7.0%)	0.842 (+7.7%)	0.853 (+6.6%)
	w/o UAF	0.865 (+5.4%)	0.829 (+9.0%)	0.872 (+4.7%)	0.909 (+1.0%)	0.857 (+6.2%)
	UDRL	0.919	0.919	0.919	0.919	0.919

Table 4

Comparison of different textual and visual backbone combinations on Weibo, Gossipcop and Politifact datasets.

Dataset	Textual backbone	Visual backbone	Accuracy	Fake news		Real news	
				Precision	F1	Precision	F1
<i>Weibo</i>	BERT	ResNet	0.920	0.918	0.926	0.923	0.919
	LSTM	Swin-T	0.893	0.890	0.895	0.896	0.890
	LSTM	ResNet	0.887	0.886	0.889	0.888	0.885
	BERT	Swin-T	0.926	0.924	0.927	0.928	0.924
<i>Gossipcop</i>	BERT	ResNet	0.879	0.793	0.685	0.896	0.925
	LSTM	Swin-T	0.864	0.753	0.643	0.886	0.916
	LSTM	ResNet	0.859	0.736	0.631	0.883	0.913
	BERT	Swin-T	0.895	0.865	0.718	0.901	0.935
<i>Politifact</i>	BERT	ResNet	0.905	0.895	0.907	0.917	0.904
	LSTM	Swin-T	0.892	0.892	0.892	0.892	0.892
	LSTM	ResNet	0.878	0.868	0.880	0.889	0.877
	BERT	Swin-T	0.919	0.919	0.919	0.919	0.919

model improves the F1 score for fake news by 1.2% and for real news by 1.1%. On the Gossipcop dataset, it delivers a 2.5% increase for fake news and 0.7% for real news. On the Politifact dataset, our model shows an improvement of 1.5% in F1 for fake news and 1.2% for real news. Moreover, compared to the GPT-4 model, our approach demonstrates superior performance. We attribute this to the fact that GPT-4 fails to capture key fake news patterns, such as tone, sentiment, and image manipulation, which are crucial for multimodal fake news detection. These findings emphasize the effectiveness of our uncertainty representation approach, as well as the decoupling and fusion of modality-private features, particularly when outperforming strong models like Miner-uv and BMR.

(3) Outperformance of Uncertainty-Based Models: Our model demonstrates significantly better results compared with uncertainty-aware model like MM-ULN. Specifically, on the Weibo dataset, our model improves the F1 score for fake news by 5.9% and for real news by 5.1%. This improvement is attributed to our advanced uncertainty-capturing modules, which more effectively capture both intra-modal and cross-modal interactions, enabling a more accurate quantification of uncertainty. Additionally, our decoupling framework allows for better separation of modality representations, leading to more precise information capture and overall enhanced performance.

(4) Robustness Across Different Datasets: Our model demonstrates strong performance across diverse datasets, excelling in both Chinese and English datasets, as well as in balanced and imbalanced datasets. On the highly imbalanced GossipCop dataset, with an approximate real-to-fake news ratio of 7:2, our model consistently performs well. For instance, the gap between F1 scores for fake news and real news in FSRU is 30.4%, while our model reduces this gap to 22.1%. This improvement is mainly due to our decoupling of modality-private features and the use of an uncertainty-aware fusion module, which adjusts the weights of different modality representations based on their certainty. This allows for more balanced and effective feature integration, leading to better detection of both real and fake news across diverse data distributions.

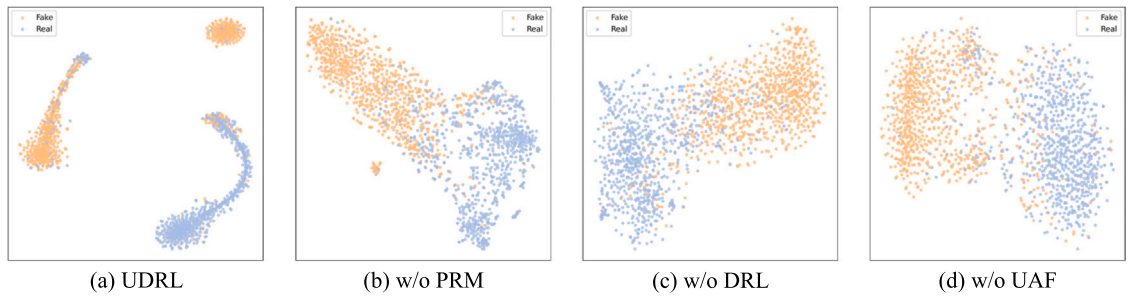


Fig. 5. T-SNE visualization of learned features before predictor by UDRL and its variants of test set on Weibo dataset.

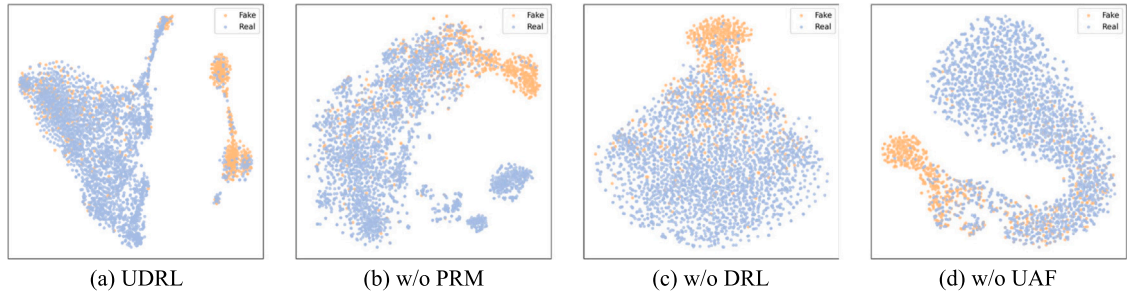


Fig. 6. T-SNE visualization of learned features before predictor by UDRL and its variants of test set on Gossipcop dataset.

4.6. Ablation study

4.6.1. Effect of each component

We conduct ablation studies by comparing the proposed UDRL with several variants to investigate the impacts of each component of UDRL, including:

- **w/o T**: The text modality is removed, and only the visual modality is used for detection.
- **w/o V**: The visual modality is removed, and only the text modality is used for detection.
- **w/o PRM**: The probability representation module is removed, and features are no longer modeled as distributions.
- **w/o DRL**: The disentangle representation learning framework is removed, and the text and visual representations are directly fused without decoupling.
- **w/o UAF**: The uncertainty-aware fusion module is replaced with simple concatenation to connect the different modality representations.

Table 3 presents the results of the ablation studies. Overall, compared to the full UDRL model, all five variants show a decrease in performance, highlighting the importance of each component. First, both UDRL w/o *T* and w/o *V* exhibit reduced effectiveness, indicating that relying solely on unimodal data is insufficient for accurate detection. Second, UDRL w/o PRM experiences a significant performance drop, underscoring the value of uncertainty modeling in visual and textual modalities to enhance multimodal representation. Third, the weaker performance of UDRL w/o DRL demonstrates the effectiveness of disentangling multimodal from unimodal features. Finally, UDRL w/o UAF results in a performance drop of up to 5.4% across the three datasets, proving that incorporating uncertainty into the fusion of modality representations significantly boosts performance.

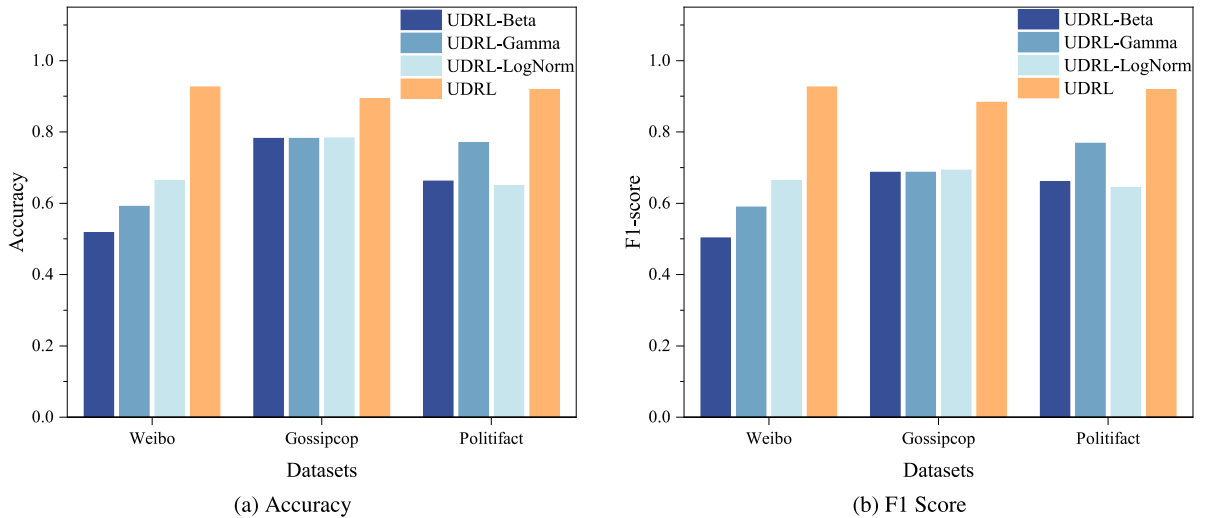
Moreover, We conducted additional experiments to compare different combinations of textual and visual feature extractors. As shown in Table 4, BERT outperforms LSTM across all datasets, demonstrating its stronger ability to model contextual dependencies. Similarly, Swin-T achieves better results than ResNet, highlighting the effectiveness of its hierarchical Transformer architecture in capturing visual patterns. These results confirm the advantage of using BERT and Swin-T as backbones for our framework.

To provide a more intuitive understanding of each component's contribution, we conducted feature visualization using T-SNE dimensionality reduction on Weibo and Gossipcop, as illustrated in Figs. 5 and 6. The visualization reveals that our complete UDRL model generates feature embeddings with clearer decision boundaries compared to all variants. Notably, without PRM, the feature distribution exhibits increased inter-class overlap, confirming that uncertainty modeling enhances representation discrimination. The absence of DRL results in less compact clusters, particularly for samples containing subtle cross-modal inconsistencies, highlighting its effectiveness in extracting complementary information. Most significantly, removing UAF leads to substantially blurred boundaries between real and fake news embeddings across all datasets, supporting our quantitative findings that uncertainty-guided fusion is critical for optimal performance. These observations provide visual evidence supporting our ablation study results in Table 3.

Table 5

Ablation study on the probability representation module with VAE and MLP substitutions.

Datasets	Models	Accuracy	Fake news		Real news	
			Precision	F1	Precision	F1
Weibo	UDRL-VAE	0.909 (+1.7%)	0.909 (+1.5%)	0.911 (+1.6%)	0.910 (+1.8%)	0.908 (+1.6%)
	UDRL-MLP	0.900 (+2.6%)	0.923 (+0.1%)	0.899 (+2.8%)	0.878 (+5.0%)	0.901 (+2.3%)
	UDRL	0.926	0.924	0.927	0.928	0.924
Gossipcop	UDRL-VAE	0.880 (+1.5%)	0.787 (+7.8%)	0.691 (+2.7%)	0.899 (+0.2%)	0.926 (+0.9%)
	UDRL-MLP	0.884 (+1.1%)	0.766 (+9.9%)	0.717 (+0.1%)	0.912 (−1.1%)	0.927 (+0.8%)
	UDRL	0.895	0.865	0.718	0.901	0.935
Politifact	UDRL-VAE	0.851 (+6.8%)	0.842 (+7.7%)	0.853 (+6.6%)	0.861 (+5.8%)	0.849 (+7.0%)
	UDRL-MLP	0.878 (+4.1%)	0.868 (+5.1%)	0.880 (+3.9%)	0.889 (+3.0%)	0.877 (+4.2%)
	UDRL	0.919	0.919	0.919	0.919	0.919

**Fig. 7.** Performance comparison of UDRL with various distribution modeling methods.

4.6.2. Superiority of probability representation module

To further demonstrate the effectiveness of the proposed Probability Representation Module, we conduct an ablation study comparing UDRL with two variants: UDRL-VAE, which replaces PRM with a Variational Autoencoder (VAE), and UDRL-MLP, which substitutes PRM with a Multi-Layer Perceptron (MLP). The results, shown in Table 5, indicate that both VAE and MLP underperform compared to PRM. While VAE models data distributions via a latent space, its simple encoder structure and reliance on KL divergence between latent distribution and standardized normal distribution reduce flexibility in capturing complex dependencies within high-dimensional information. Similarly, MLP, though effective for basic discriminative tasks, lacks the probabilistic modeling capacity to represent information uncertainty and intricate relationships. In contrast, proposed PRM, designed to capture richer distributional information and complex dependencies, consistently achieves superior performance across tasks.

In addition to these structure variations, we explore various distribution modeling methods, including Beta distribution, Gamma distribution, and Log-Normal distribution. The experimental results in Fig. 7 show significant drops in both accuracy and F1 scores when these alternative distributions are used, demonstrating the robustness and versatility of the Gaussian distribution. The superior performance of Gaussian distribution may be attributed to several factors: (1) its unbounded support is particularly suitable for modeling feature representations in deep neural networks which often span the entire real number space; and (2) its symmetrical nature aligns well with the error distributions typically observed in representation learning. In contrast, Beta distribution's bounded support $[0,1]$ introduces artificial constraints on the representation space, while Log-Normal and Gamma distributions impose asymmetry that may not accurately reflect the underlying uncertainty characteristics of our multimodal representations. Unlike the alternatives, which may introduce constraints or bias in certain data contexts, the Gaussian distribution proves highly adaptable across different feature distributions, leading to more consistent and reliable performance in our model.

4.6.3. Analysis of disentangle representation learning

To evaluate the contribution of each constraint in our disentangled representation learning framework, we conducted ablation studies and examined the results through both performance metrics and feature space visualization.

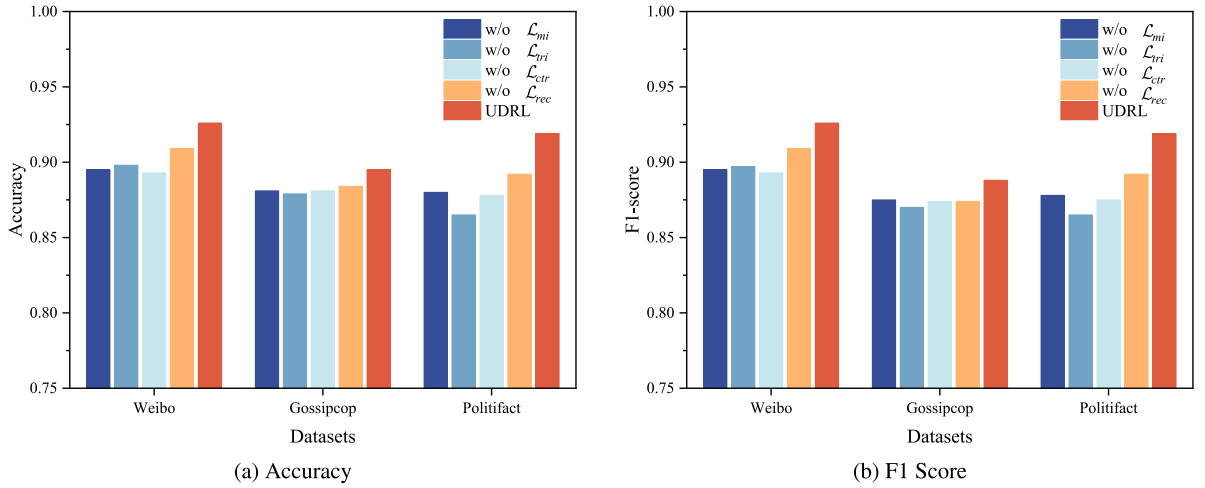


Fig. 8. Ablation results of different loss in UDRL on three datasets.

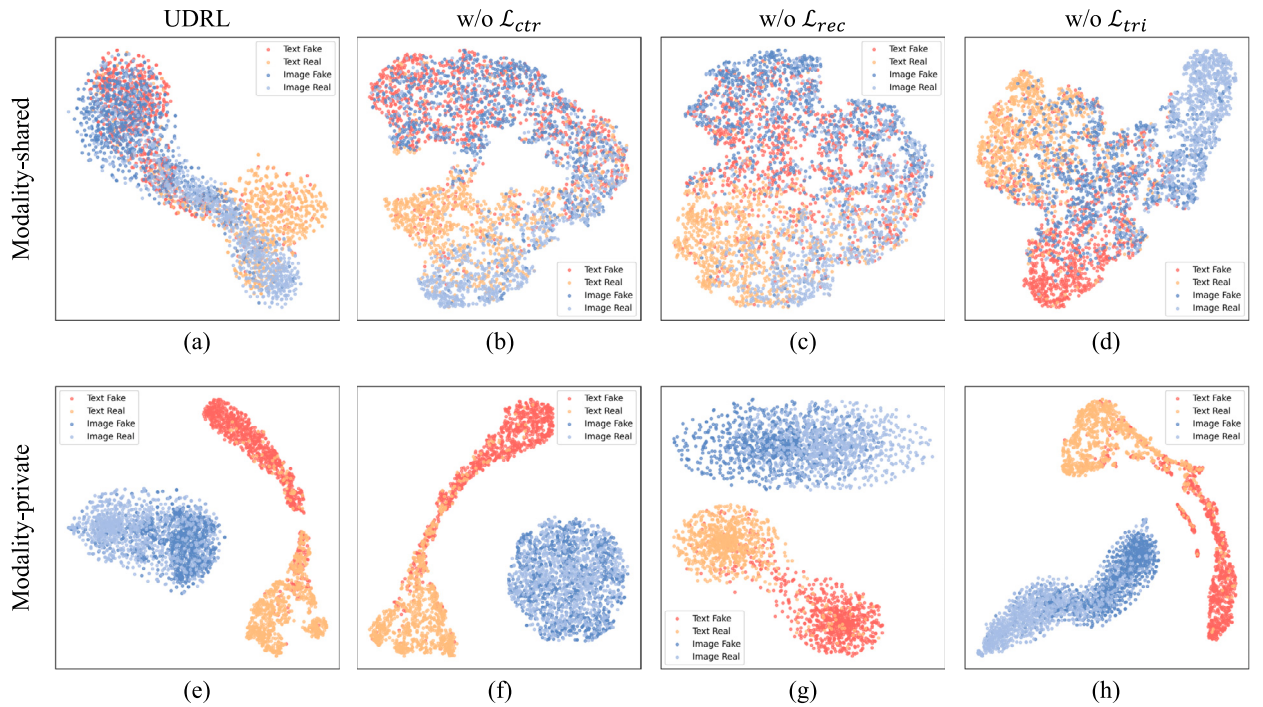


Fig. 9. T-SNE visualization of learned disentangled representations by UDRL and its variants of the test set on the Weibo dataset. The red points represent Text Fake, the orange points represent Text Real, the dark blue points represent Image Fake, and the light blue points represent Image Real. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

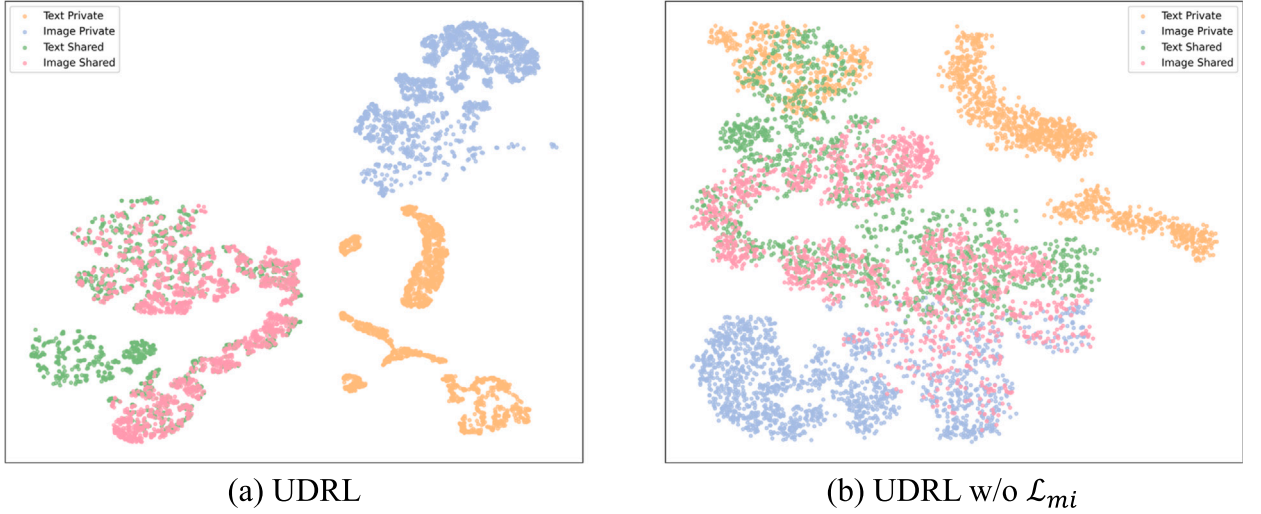


Fig. 10. T-SNE visualization of learned disentangled representations by UDRL and UDRL w/o $\mathcal{L}_{mi}$ of test set on Weibo dataset. The orange points represent Text Private, the green points represent Text Shared, the blue points represent Image Private, and the pink points represent Image Shared. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

Performance Impact Analysis. As illustrated in Fig. 8, removing any individual constraint leads to decreased performance across all datasets, confirming each component’s value. The mutual information loss ($\mathcal{L}_{mi}$) maintains clear boundaries between modality-unique and shared content; the triplet loss ($\mathcal{L}_{tri}$) bridges text and image shared representations; the center loss ($\mathcal{L}_{ctr}$) sharpens within-modality feature distinctions; and the reconstruction loss ($\mathcal{L}_{rec}$) preserves information integrity. Interestingly, omitting $\mathcal{L}_{rec}$ caused the smallest performance decline, suggesting that the remaining objectives adequately maintain essential task information even without explicit reconstruction guidance.

Feature Space Visualization Analysis. To complement our numerical findings, we employed T-SNE to visualize how each constraint shapes the learned representation space, as shown in Figs. 9 and 10:

- **Complete Model Analysis:** Our full model creates a representation space where modality-private and shared features occupy distinct regions. Shared features from both text and images naturally cluster by content rather than by their source modality, with clear boundaries separating fake and real news categories.
- **Impact of Mutual Information Loss:** Without $\mathcal{L}_{mi}$ (Fig. 10), the distinction between modality-private and shared information dissolves. The visualization reveals how this constraint acts as a gatekeeper, preventing modality-unique characteristics from contaminating the shared semantic space.
- **Impact of Triplet Loss:** Fig. 9(d) shows that eliminating $\mathcal{L}_{tri}$ causes shared representations to segregate by modality rather than content. This demonstrates how triplet loss serves as a bridge between modalities, enabling cross-modal understanding through aligned representations.
- **Impact of Center Loss:** The absence of $\mathcal{L}_{ctr}$ (Fig. 9(b)) results in scattered, less defined class groupings. This visualization highlights how center loss acts as a gravitational force, pulling same-class representations together to form more coherent, discriminative clusters.
- **Impact of Reconstruction Loss:** Fig. 9(c) reveals that without $\mathcal{L}_{rec}$, the representation structure largely persists with only subtle changes. This visual evidence supports our quantitative finding that reconstruction loss, while beneficial, plays a complementary role to the other constraints that maintain the core representational architecture.

Through this dual-perspective analysis, we gain deeper insights into how each component contributes to our framework’s effectiveness. The performance metrics quantify the impact, while the visualizations illuminate the mechanisms through which each constraint shapes the representation space. Together, they demonstrate how our approach successfully balances modality-private information extraction with cross-modal alignment, creating representations that capture the essential characteristics needed for effective fake news detection.

4.6.4. Superiority of uncertainty-aware fusion module

We evaluate alternative approaches for fusing modality-spec and modality-shared features, including UDRL UDRL w/o U (uncertainty reweighting is removed), UDRL-SA (self-attention mechanism is used to fuse features) and UDRL-Concat(concatenation is used to fuse features). Table 6 displays the experimental results, which clearly demonstrate the superiority of UDRL over its variants. UDRL not only effectively incorporates uncertainty reweighting but also utilizes a cross-attention mechanism to capture the complex interactions between modalities. This dual capability allows UDRL to excel in performance, achieving the highest accuracy and F1 scores across all datasets.

Table 6

Ablation study of different fusion methods in UDRL on Weibo, Gossipcop and Politifact datasets.

Datasets	Models	Accuracy	Fake news		Real news	
			Precision	F1	Precision	F1
Weibo	UDRL w/o U	0.913 (+1.3%)	0.933 (−0.9%)	0.913 (+1.4%)	0.894 (+3.4%)	0.914 (+1.0%)
	UDRL-SA	0.909 (+1.7%)	0.909 (+1.5%)	0.911 (+1.6%)	0.910 (+1.8%)	0.908 (+1.6%)
	UDRL-Concat	0.898 (+2.8%)	0.903 (+2.1%)	0.899 (+2.8%)	0.893 (+3.5%)	0.897 (+2.3%)
	UDRL	0.926	0.924	0.927	0.928	0.924
Gossipcop	UDRL w/o U	0.887 (+0.8%)	0.807 (+5.8%)	0.689 (+2.9%)	0.896 (+0.5%)	0.927 (+0.8%)
	UDRL-SA	0.882 (+1.3%)	0.791 (+7.4%)	0.691 (+2.7%)	0.899 (+0.2%)	0.926 (+0.9%)
	UDRL-Concat	0.880 (+1.5%)	0.777 (+8.8%)	0.697 (+1.5%)	0.902 (−0.1%)	0.925 (+1.0%)
	UDRL	0.895	0.865	0.718	0.901	0.935
Politifact	UDRL w/o U	0.892 (+2.7%)	0.892 (+2.7%)	0.892 (+2.7%)	0.892 (+2.7%)	0.892 (+2.7%)
	UDRL-SA	0.878 (+4.1%)	0.818 (+10.1%)	0.889 (+3.0%)	0.967 (−4.8%)	0.866 (+5.3%)
	UDRL-Concat	0.865 (+5.4%)	0.865 (+5.4%)	0.865 (+5.4%)	0.865 (+5.4%)	0.865 (+5.4%)
	UDRL	0.919	0.919	0.919	0.919	0.919

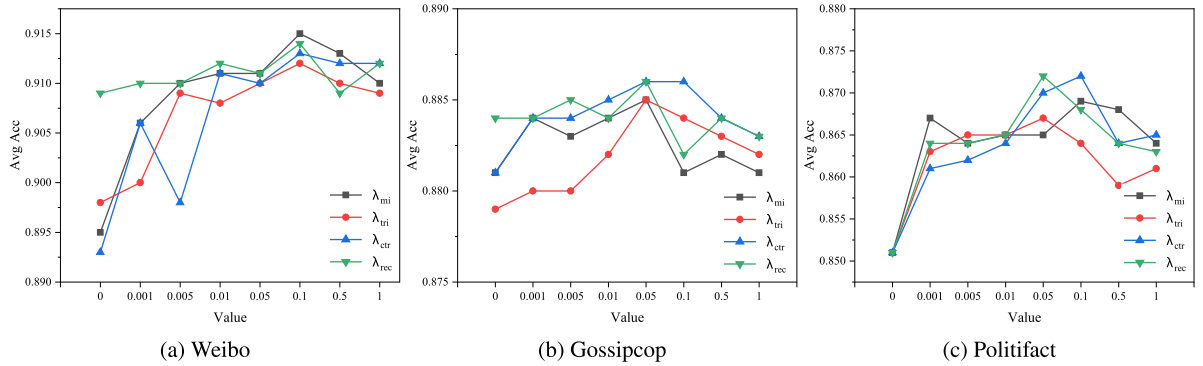
**Fig. 11.** Sensitivity analysis of λ_{mi} , λ_{tri} , λ_{ctr} , λ_{rec} in UDRL on three datasets.

Image	Text	Label	Prediction	$D_{KL}(z_p^u z_s)$	$D_{KL}(z_p^t z_s)$
	Frequent flyers: The 20 yuan aviation accident insurance also covers flight delays—2,000 yuan for delays over 4 hours and 3,000 yuan for over 8 hours. Many passengers miss out on these benefits, making it worth sharing	Fake	Fake		Frequent flyers: The 20 yuan aviation accident insurance also covers flight delays—2,000 yuan for delays over 4 hours and 3,000 yuan for over 8 hours. Many passengers miss out on these benefits, making it worth sharing
	A pig-human hybrid was discovered on Lingshan Tower in Shuangyang Town, Zhangping City, Fujian Province. Is this news true	Fake	Fake		A pig-human hybrid was discovered on Lingshan Tower in Shuangyang Town, Zhangping City, Fujian Province. #Is this news true
	Breaking: FBI Issues Warrant for Obama's Arrest After Confirming Illegal Trump Tower Wiretap	Fake	Fake		Breaking: FBI Issues Warrant for Obama's Arrest After Confirming Illegal Trump Tower Wiretap
	More Than Just Shock. Before being swept away by the flood, an Indonesian man calmly posed for the camera, making his final gesture. Perhaps it was a sign of respect or a way to preserve beauty beyond fear. Each time the principal sees this photo, they feel deeply moved. Cherish everything in front of you!	Fake	Fake		More Than Just Shock. Before being swept away by the flood, an Indonesian man calmly posed for the camera, making his final gesture. Perhaps it was a sign of respect or a way to preserve beauty beyond fear. Each time the principal sees this photo, they feel deeply moved. Cherish everything in front of you!

Fig. 12. The KL divergence visualization and prediction of our UDRL. For visual modality, a higher KL divergence is represented by brighter colors (yellow) while a higher KL divergence for the textual modality is represented by darker shades.

4.7. Sensitivity analysis

Following our previous ablation study on the four loss constraints, we conducted a sensitivity analysis on each loss parameter through a grid search within the range [0,1], as shown in Fig. 11. The findings reveal two primary insights: First, performance

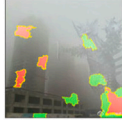
Image	Text	Label	Prediction	Explanation (V)	Explanation (T)
	The city is engulfed in heavy smog, with an orange alert issued. Some areas have off-the-charts pollution levels, worsening tonight. Stay indoors if possible, avoid driving, and wear multiple masks. #BeijingSmog	True	Fake		The city is engulfed in heavy smog , with an orange alert issued. Some areas have off-the-charts pollution levels, worsening tonight. Stay indoors if possible, avoid driving, and wear multiple masks. #BeijingSmog
	Will You Share to Save Them? Yichang police seized 7 macaques from an illegally parked vehicle. The suspects confessed that the monkeys were to be sold for use in performances. Baby monkeys were beaten with whips and sticks—can you imagine their suffering? Don't find joy in their pain. Reject animal performances, reject wildlife trade, save them! Share if you support!	True	Fake		Will You Share to Save Them? Yichang police seized 7 macaques from an illegally parked vehicle. The suspects confessed that the monkeys were to be sold for use in performances. Baby monkeys were beaten with whips and sticks—can you imagine their suffering ? Don't find joy in their pain. Reject animal performances, reject wildlife trade, save them! Share if you support!
	Alabama's Crimson Tide was disqualified from national championship.	Fake	True		Alabama's Crimson Tide was disqualified from national championship .
	Piranhas in Liuzhou, Guangxi Identified as Released by Japanese Nationals. Hirahara Ichiro, 45, and Noda Yoshihide, 22, admitted to raising piranhas at their residence and releasing them into the Liuijiang River. Evidence confirms their actions. Both admitted to releasing piranhas into the Liuijiang River. The incident is under investigation.	Fake	True		Piranhas in Liuzhou, Guangxi Identified as Released by Japanese Nationals. Hirahara Ichiro, 45, and Noda Yoshihide, 22, admitted to raising piranhas at their residence and releasing them into the Liuijiang River. Evidence confirms their actions. Both admitted to releasing piranhas into the Liuijiang River. The incident is under investigation .

Fig. 13. Error case and LIME explanation of UDRL on Weibo, Gossipcop and Politifact datasets. For visual modality, green areas indicate the regions that positively support the model's decision, while red areas indicate regions that contradict it. For text modality, words highlighted in orange represents terms with significant influence on model's decision.

improves consistently across all cases when parameters are non-zero, highlighting the necessity of each constraint parameter in enhancing model performance. Second, accuracy reaches its peak when the parameters lie within the range [0.05, 0.1], suggesting this as an optimal range. This range appears to strike a balance, where each constraint contributes meaningfully to disentangled representation learning without overburdening the model's capacity.

4.8. Case study

Good case. To demonstrate the effectiveness of our model and enhance interpretability, we select four representative cases and visualize the contributions of modality-private and modality-shared features, as shown in Fig. 12. In the visualization, a higher KL divergence for the visual modality is represented by brighter colors, while a higher KL divergence for the textual modality is represented by darker shades.

For Case 1, the text discusses aviation accident insurance, but the information is fabricated, making the news fake. The image, showing a child's face, has weak alignment with the content. Both the visual and textual KL divergence are high, indicating that the model relies on both modality-private features to detect the deception. For Case 2, the claim about a pig-human hybrid is paired with a manipulated image. Since the image is heavily edited, the visual KL divergence is high, leading the model to focus more on the visual-private features for detection. For Case 3, the fake news about an FBI warrant for Obama's arrest is supported by a manipulated image. The visual KL divergence is high due to the image's inconsistency with the actual event, so the model pays more attention to the image's private features. For Case 4, the text describes an emotional flood story, while the image is aligned with the narrative. However, the text contains vague terms like 'perhaps' and emotional cues like 'moved', leading to a higher KL divergence for the textual modality. The model thus places more weight on the text-private features for detection. These cases highlight how our model dynamically adjusts the importance of modality-private and shared features, effectively capturing deception patterns in multimodal fake news detection.

Bad case. To better understand our model's limitations and decision-making process, we conducted a detailed error case analysis using LIME[†] (Local Interpretable Model-agnostic Explanations). LIME approximates our complex model locally with an interpretable model around private predictions by perturbing input features and observing prediction changes, creating a local linear model that highlights which features most strongly influence classification decisions.

Our analysis revealed several interesting patterns across both false positives and false negatives, as shown in Fig. 13. In the visualizations, green areas in images indicate regions that positively support the model's decision, while red areas indicate regions that contradict it. For text, words highlighted in orange represent terms with significant influence on the model's decision.

False Positives (Real News Classified as Fake): We identified two representative cases where our model incorrectly flagged authentic news as fake. In the first case, a genuine news article about severe smog conditions was misclassified primarily due to its

[†] <https://github.com/marcotcr/lime>.

Table 7

Performance of UDRL and Miner-uvs on real-world Chinese and English multimodal fake news datasets.

Dataset	Model	Accuracy	Fake news			Real news		
			Precision	Recall	F1	Precision	Recall	F1
Chinese	Miner-uvs	0.820	0.808	0.840	0.824	0.833	0.800	0.816
	UDRL	0.840	0.854	0.820	0.837	0.827	0.860	0.843
English	Miner-uvs	0.760	0.796	0.700	0.745	0.732	0.800	0.774
	UDRL	0.790	0.837	0.720	0.774	0.764	0.860	0.804

accompanying image. The image, while related to the topic, had been digitally altered to include a monster-like figure emerging from the smog. LIME analysis revealed that the model heavily weighted visual inconsistencies while undervaluing the factual accuracy of the textual content. In the second case, a news report about monkeys being captured was misclassified as fake. LIME analysis showed that the model focused on emotionally provocative language such as “please share this widely” and rhetorical questions expressing strong emotions. The model identified these as sensationalist features commonly associated with fake news, despite the content being factual.

False Negatives (Fake News Classified as Real): Our analysis identified sophisticated fake news designed with high cross-modal coherence posed significant challenges. In one representative case, a fabricated story claiming “Alabama Crimson Tide Football Team Disqualified from National Championship” demonstrated exceptional text-image consistency, with professionally created imagery matching the false narrative. LIME highlighted that both text and image modalities produced low uncertainty signals, with neither containing obvious manipulation markers. In another case, a fabricated news story about a foreigner deliberately releasing piranhas in Chinese waters included highly detailed but entirely fictional biographical information about the perpetrator. The specificity and coherence of these details created an illusion of credibility that our model incorrectly interpreted as indicators of authenticity.

Based on this error analysis, we identify several promising future directions, including incorporating external knowledge sources, leveraging large language models for contextual understanding, and enhancing our framework with social context information to better address these challenging cases.

4.9. Real world scenarios performance

To comprehensively evaluate the performance of our proposed UDRL model in real-world scenarios, we constructed two multimodal fake news datasets: one in Chinese and one in English. Each dataset contains 100 samples covering nine different domains, with each sample including both textual and visual modalities.

Dataset Construction Methodology. We adopted different approaches for collecting real news and fake news to ensure data quality and authenticity. For real news, we primarily obtained content from official accounts of authoritative media on major social media platforms. In the Chinese context, these included verified accounts from People’s Daily and Xinhua News Agency on Weibo, along with official media accounts on WeChat and Zhihu. For English content, we gathered from verified accounts of CNN, BBC, Reuters and other mainstream media on X. For fake news collection, we employed a different methodology focused on professional fact-checking resources. The Chinese fake news samples were collected from specialized debunking platforms including the China Internet Joint Rumor Debunking Platform,[‡] (directed by the Cyberspace Administration of China), Toutiao Fact Check,[§] and Zhihu’s Rumor Refutation Section[¶]. The English fake news samples were sourced from established fact-checking organizations including Snopes,^{||} and FactCheck^{**}. During collection, we ensured we obtained the original false information (including both text and images) and its dissemination status, recorded the fact-checking results provided by the platforms, and prioritized fake news cases with significant social impact.

To ensure the diversity and representativeness of the dataset, we distributed the collected samples evenly across nine different domains: Finance, Health, Military, Science, Politics, International Relations, Education, Entertainment, and Society. Each domain contains approximately 10–12 samples, with about 5–6 real news and 5–6 fake news items, ensuring the balance of the dataset.

Experimental Setup. To evaluate the effectiveness of our UDRL model on these real-world datasets, we utilized our pre-trained models and the competitive model Miner-uvs pretrained from benchmark experiments. Specifically, we used the model trained on the Weibo dataset for testing the Chinese samples and the model trained on GossipCop (which contains more data compared to PolitiFact) for testing the English samples. This approach allows us to assess how well our models, trained on larger benchmark datasets, generalize to new, diverse real-world examples without additional fine-tuning. We maintained the same model architecture and parameters as described in the benchmark experiments section.

Results and Analysis. Table 7 presents the performance of our UDRL model on the constructed Chinese and English datasets. The model achieved an accuracy of 84.0% on the Chinese dataset when transferring knowledge from the Weibo benchmark, with

[‡] <https://www.piyao.org.cn>.

[§] <https://www.toutiao.com>.

[¶] https://www.zhihu.com/appview/community_construction.

^{||} <https://www.snopes.com>.

^{**} <https://www.factcheck.org>.

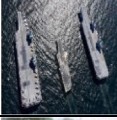
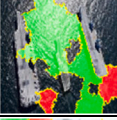
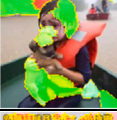
Image	Text	Label	Prediction	Explanation (V)	Explanation (T)
	An elderly woman was fined 160,000 yuan for selling fruit at a street stall. She found the fine amount outrageous and refused to pay. The law enforcement officers then threatened that if she didn't pay, the fine would increase to 1.45 million yuan.	Fake	Fake		An elderly woman was fined 160,000 yuan for selling fruit at a street stall. She found the fine amount outrageous and refused to pay. The law enforcement officers then threatened that if she didn't pay, the fine would increase to 1.45 million yuan.
	HMS Queen Elizabeth (left) and HMS Prince of Wales (right) have rendezvoused with French aircraft carrier Charles de Gaulle (centre) in the North Sea today.	Fake	Fake		HMS Queen Elizabeth (left) and HMS Prince of Wales (right) have rendezvoused with French aircraft carrier Charles de Gaulle (centre) in the North Sea today.
	This picture has been seared into my mind. My heart hurts.	Fake	True		This picture has been seared into my mind. My heart hurts .
	Earthquake in Baicheng County, Xinjiang: 3 Deaths and 65 Houses Damaged	Fake	True		Earthquake in Baicheng County, Xinjiang: 3 Deaths and 65 Houses Damaged

Fig. 14. Case study of UDRL on real world scenarios.

balanced performance in detecting both fake news ($F1 = 83.7\%$) and real news ($F1 = 84.3\%$). On the English dataset, our model achieved an accuracy of 79.0% when transferring from the GossipCop benchmark, with a notable gap between fake news detection ($F1 = 77.4\%$) and real news classification ($F1 = 80.4\%$), which is consistent with the performance on the GossipCop dataset. Compared to Miner-uvs, which achieves an accuracy of 76.0% on the English dataset, our model surpasses it by 3.0%, demonstrating the effectiveness of our disentangle representation learning framework and uncertainty-aware fusion mechanism. The results further highlight effective knowledge transfer across datasets, with stronger performance on the Chinese dataset compared to the English dataset.

Case Study. To gain deeper insights into the model's decision-making process, we analyzed four representative examples from our datasets using LIME for decision visualization, shown in Fig. 14. For the first case, the model correctly identified fake news by focusing on emotionally charged and exaggerated phrases such as “outrageous” and “threatened”, recognizing patterns typical of sensationalist content. For the second case, the model successfully detected manipulation by identifying visual inconsistencies in the image, particularly at the boundaries where unnatural elements appeared, demonstrating its ability to detect photoshopped images. For the third case, the model misclassified the content as real, focusing on emotionally consistent terms like “seared” and “hurts” that aligned well with the image, failing to detect the AI-generated nature of the visual. For the fourth case, the model incorrectly classified the content as real, focusing on the semantic coherence between text and image while failing to detect that the image was repurposed from a different event. These findings highlight the model's strengths in detecting exaggerated textual content and obvious visual manipulations, while revealing limitations in identifying sophisticated AI-generated images and contextually misused authentic images.

5. Conclusion

In this paper, we proposed the Uncertainty-aware Disentangled Representation Learning framework for multimodal fake news detection, addressing critical challenges in capturing uncertainty, managing modality heterogeneity, and balancing attention between modality-private veracity and cross-modal inconsistencies. Our approach leverages probabilistic representation to model multimodal data as Gaussian distributions, effectively capturing the inherent ambiguity in multimodal information. The disentangled representation learning framework separates shared and private information across modalities, resulting in robust and discriminative representations. Additionally, our uncertainty-aware fusion module dynamically adjusts modality importance based on uncertainty, facilitating more precise cross-modal interactions. Experimental results on three benchmark datasets demonstrate that UDRL achieves consistently strong performance compared to recent competitive methods.

Future research could focus on enhancing the interpretability of fake news detection by integrating explainable AI techniques, as well as incorporating knowledge graphs or large language models to provide richer contextual understanding and improve detection accuracy. These extensions could further advance UDRL's adaptability and effectiveness in real-world, complex multimodal scenarios.

CRedit authorship contribution statement

Zihao Li: Writing – original draft, Validation, Methodology, Investigation, Formal analysis, Data curation, Conceptualization.
Jiaxin Yang: Writing – review & editing, Validation, Methodology, Investigation, Formal analysis, Data curation, Conceptualization.

Xianghan Wang: Writing – review & editing, Validation, Investigation. **Jun Lei:** Writing – review & editing, Resources, Funding acquisition. **Shuohao Li:** Resources, Funding acquisition. **Jun Zhang:** Writing – review & editing, Supervision, Project administration, Funding acquisition, Conceptualization.

Ethical considerations

Our research on automated fake news detection raises several important ethical implications:

- **Model Biases and Fairness:** We identified potential imbalances across news domains and cultural-linguistic nuances that may affect detection accuracy, with preliminary analyses showing performance variations requiring further investigation.
- **Classification Errors:** We acknowledge dual risks of false positives (silencing valid information) and false negatives (allowing misinformation to spread), emphasizing our model should support rather than replace human judgment.
- **Public Benefit:** Our research aims to reduce harmful misinformation without generating controversial content, contributing to healthier information ecosystems while balancing against over-moderation risks.
- **Data Ethics:** We utilized only publicly available datasets (Weibo, GossipCop and Politifact) while acknowledging their specific temporal and cultural limitations.

We advocate for responsible implementation with appropriate human oversight, continuous bias evaluation, and context-sensitive application.

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Data availability

Data will be made available on request.

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